

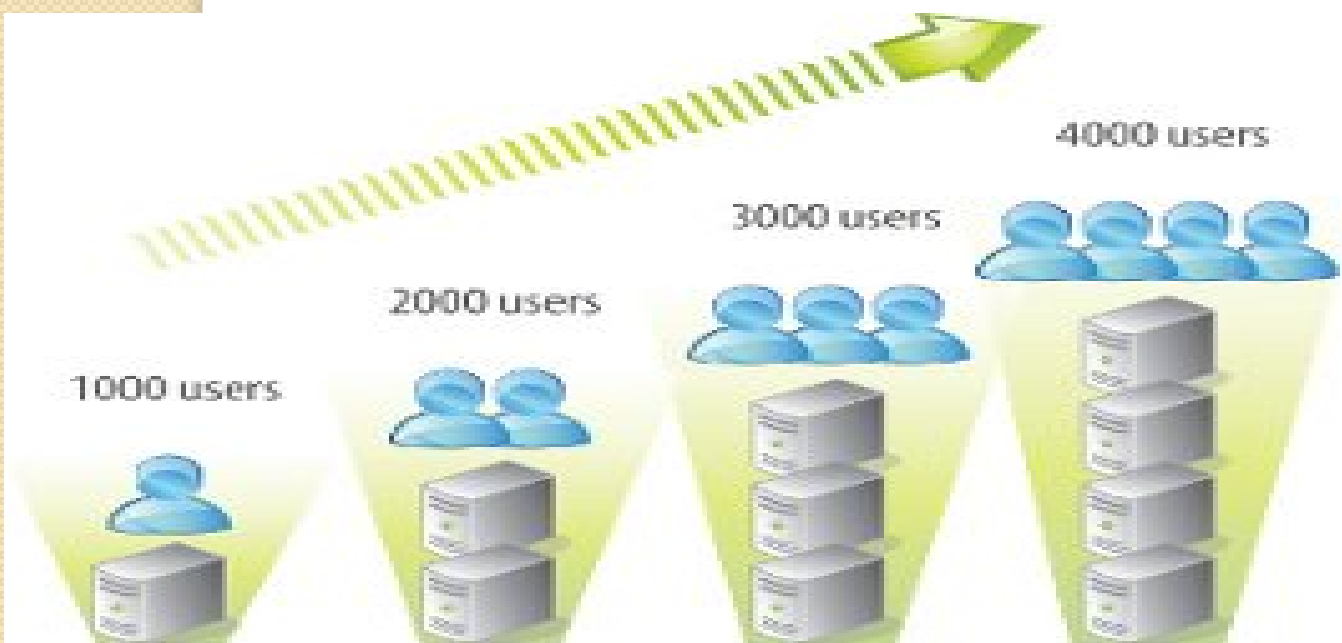
Introduction to Mobile Communication

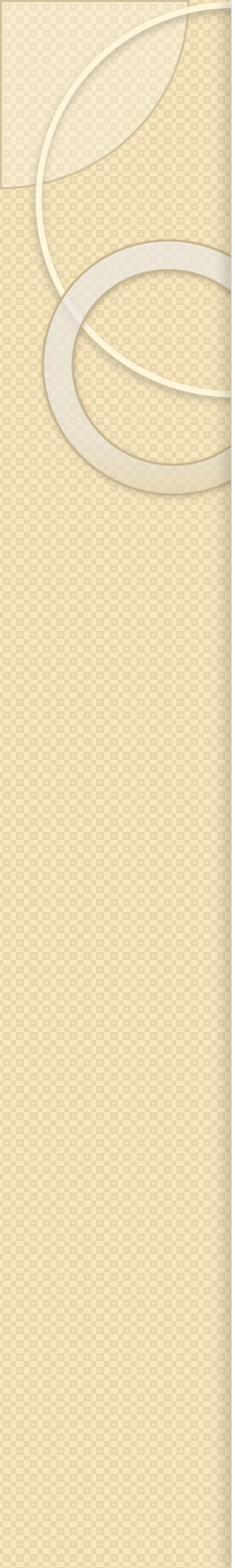
- **Mobile Communication** is the use of technology that allows us to communicate with others in different locations without the use of any physical connection (wires or cables).
- Mobile communication makes our life easier, and it saves time and effort.
- **A mobile phone** (also called mobile cellular network, cell phone or hand phone) is an example of mobile communication (wireless communication).

Features of Mobile Communication

- **High capacity load balancing:** Each wired or wireless infrastructure must incorporate high capacity load balancing. High capacity load balancing means, when one access point is overloaded, the system will actively shift users from one access point to another depending on the capacity which is available.

- **Scalability:** The growth in popularity of new wireless devices continuously increasing day by day. The wireless networks have the ability to start small if necessary, but expand in terms of coverage and capacity as needed - without having to overhaul or build an entirely new network.



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- **Network management system:** Now a day, wireless networks are much more complex and may consist of hundreds or even thousands of access points, firewalls, switches, managed power and various other components.
 - The wireless networks have a smarter way of managing the entire network from a centralized point.

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- **Indoor as well as outdoor coverage options:** It is important that your wireless system has the capability of adding indoor coverage as well as outdoor coverage.
 - **Network access control:** Network access control can also be called as mobile device registration. It is essential to have a secure registration.
Network access control (NAC) controls the role of the user and enforces policies. NAC can allow your users to register themselves to the network. It is a helpful feature that enhances the user experience.



● **Roaming** allows your end-users to successfully move from one access point to another without ever noticing a dip in a performance.

For example, allowing a student to check their mail as they walk from one class to the next.

● **Switching:** Basically, a network switch is the traffic cop of the wireless network

Switching is an essential part of every fast, secure wireless network for several reasons:

- It helps the traffic on your network flow more efficiently.
- It minimizes unnecessary traffic.
- It creates a better user experience by ensuring your traffic is going to the right places.

Advantages of Mobile Communication

- **Flexibility:** Wireless communication enables the people to communicate with each other regardless of location. There is no need to be in an office or some telephone booth in order to pass and receive messages.

- **Cost effectiveness:** In wireless communication, there is no need of any physical infrastructure (Wires or cables) or maintenance practice. Hence, the cost is reduced.

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- **Speed:** Improvements can also be seen in speed. The network connectivity or the accessibility was much improved in accuracy and speed.
 - **Accessibility:** With the help of wireless technology easy accessibility to the remote areas is possible. For example, in rural areas, online education is now possible. Educators or students no longer need to travel to far-flung areas to teach their lessons.

Multiplexing

- Access methods are multiplexing techniques that provide communications services to multiple users in a single-bandwidth wired or wireless medium.
- Communications channels, whether they're wireless spectrum segments or cable connections, are expensive.
- Communications services providers must engage multiple paid users over limited resources to make a profit.
- Access methods allow many users to share these limited channels to provide the economy of scale necessary for a successful communications business

FDMA

- FDMA is the process of dividing one channel or bandwidth into multiple individual bands, each for use by a single user .
- Each individual band or channel is wide enough to accommodate the signal spectra of the transmissions to be propagated.
- The data to be transmitted is modulated on to each subcarrier, and all of them are linearly mixed together.

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- The best example of this is the cable television system.
 - The medium is a single coax cable that is used to broadcast hundreds of channels of video/audio programming to homes.
 - The coax cable has a useful bandwidth from about 4 MHz to 1 GHz. This bandwidth is divided up into 6-MHz wide channels.

TDMA

- TDMA is a digital technique that divides a single channel or band into time slots. Each time slot is used to transmit one byte or another digital segment of each signal in sequential serial data format. This technique works well with slow voice data signals, but it's also useful for compressed video and other high-speed data.

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- A good example is the widely used T1 transmission system, which has been used for years in the telecom industry.
 - T1 lines carry up to 24 individual voice telephone calls on a single line,
 - The basic GSM (Global System of Mobile Communications) cellular phone system is TDMA-based.
 - It divides up the radio spectrum into 200-kHz bands and then uses time division techniques to put eight voice calls into one channel.

CDMA

- CDMA is another pure digital technique. It is also known as spread spectrum because it takes the digitized version of an analog signal and spreads it out over a wider bandwidth at a lower power level. This method is called direct sequence spread spectrum (DSSS) as well
- The third generation (3G) cell-phone technology called wideband CDMA (WCDMA) uses a similar method with compressed voice and 3.84-Mbit/s chipping codes in a 5-MHz channel to allow multiple users to share the same band.

DMA	TDMA	CDMA
FDMA stands for Frequency Division Multiple Access.	TDMA stands for Time Division Multiple Access.	CDMA stands for Code Division Multiple Access.
In this, sharing of bandwidth among different stations takes place.	In this, only the sharing of time of satellite transponder takes place.	In this, there is sharing of both i.e. bandwidth and time among different stations takes place.
There is no need of any codeword.	There is no need of any codeword.	Codeword is necessary.
In this, there is only need of guard bands between the adjacent channels are necessary.	In this, guard time of the adjacent slots are necessary.	In this, both guard bands and guard time are necessary.
Synchronization is not required.	Synchronization is required.	Synchronization is not required.
The rate of data is low.	The rate of data is medium.	The rate of data is high.
Mode of data transfer is continuous signal.	Mode of data transfer is signal in bursts.	Mode of data transfer is digital signal.

GSM

- GSM stands for **Global System for Mobile Communication**.
- GSM is an open and digital cellular technology used for mobile communication.
- It uses 4 different frequency bands of 850 MHz, 900 MHz, 1800 MHz and 1900 MHz . It uses the combination of FDMA and TDMA.

Features of GSM are :

- Supports international roaming
- Clear voice clarity
- Ability to support multiple handheld devices.
- Spectral / frequency efficiency
- Low powered handheld devices.
- Ease of accessing network
- International ISDN compatibility.

GSM - Architecture

- A GSM network comprises of many functional units. These functions and interfaces are explained in this chapter. The GSM network can be broadly divided into –
- The Mobile Station (MS)
- The Base Station Subsystem (BSS)
- The Network Switching Subsystem (NSS)
- The Operation Support Subsystem (OSS)

GSM - The Mobile Station

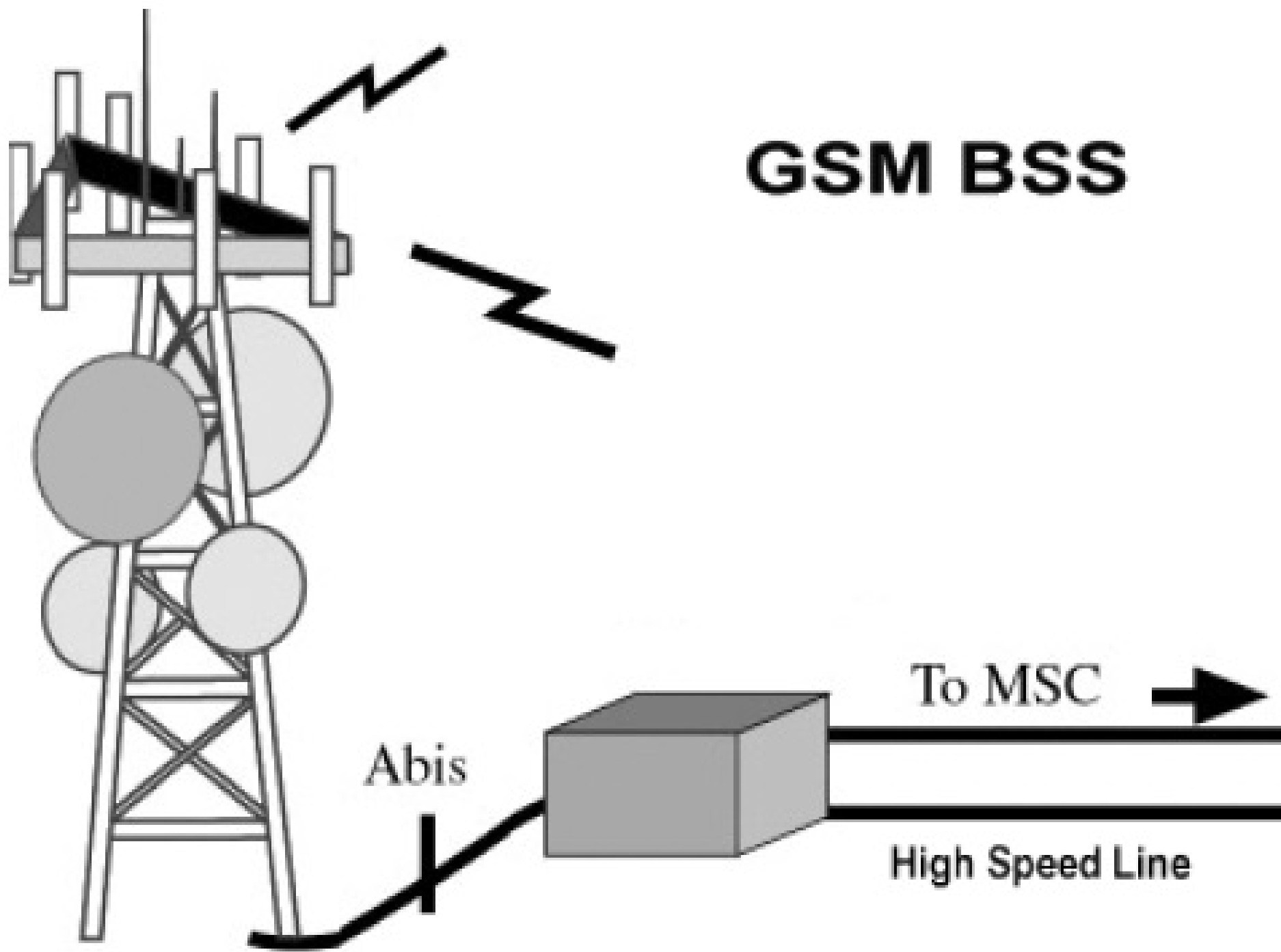
- The MS consists of the physical equipment, such as the radio transceiver, display and digital signal processors, and the SIM card.
- It provides the air interface to the user in GSM networks. As such, other services are also provided, which include –
- Voice teleservices
- Data bearer services
- The features' supplementary services

What is SIM?

- The SIM provides personal mobility so that the user can have access to all subscribed services irrespective of both the location of the terminal and the use of a specific terminal.
- It needs to insert the SIM card into another GSM cellular phone to receive calls at that phone, make calls from that phone, or receive other subscribed services.

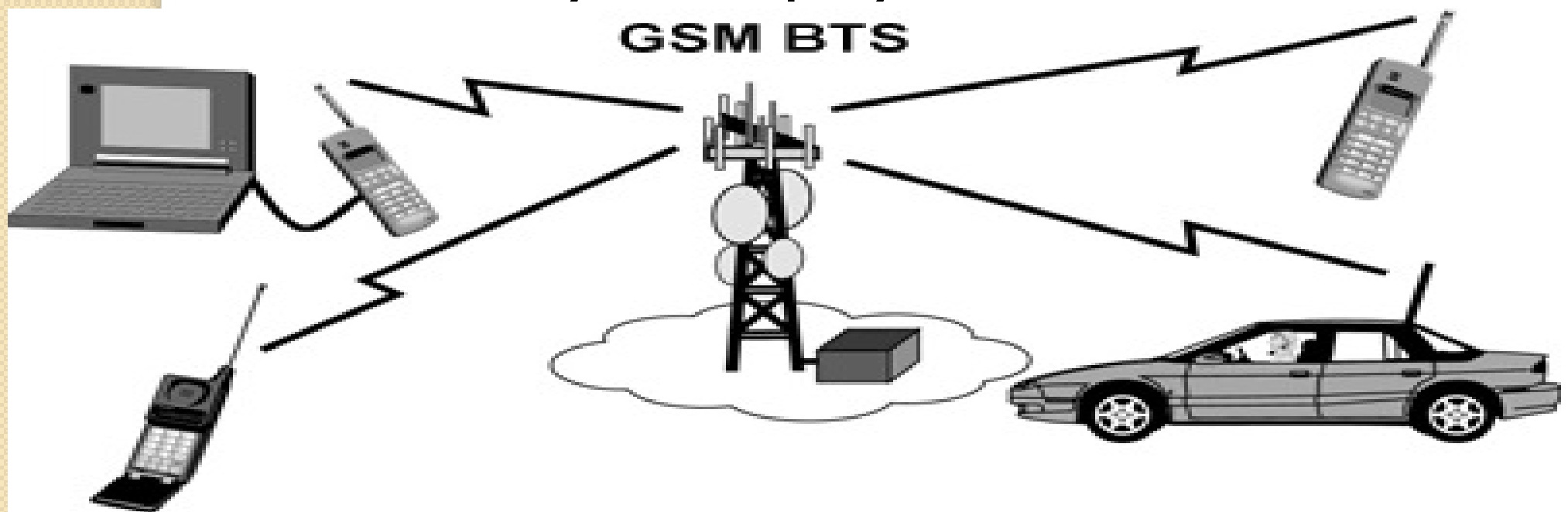
GSM - The Base Station Subsystem (BSS)

- The BSS is composed of two parts –
- The Base Transceiver Station (BTS)
- The Base Station Controller (BSC)
- The BTS and the BSC communicate across the specified Abis interface, enabling operations between components that are made by different suppliers.
- The radio components of a BSS may consist of four to seven or nine cells. A BSS may have one or more base stations. The BSS uses the Abis interface between the BTS and the BSC.



The Base Transceiver Station (BTS)

- The BTS houses the radio transceivers that define a cell and handles the radio link protocols with the MS.
- In a large urban area, a large number of BTSs may be deployed.



Each BTS serves as a single cell. It also includes the following functions –

- Encoding, encrypting, multiplexing, modulating, and feeding the RF signals to the antenna

- Transcoding and rate adaptation

- Time and frequency synchronizing

- Voice through full- or half-rate services

- Decoding, decrypting, and equalizing received signals

- Random access detection

- Timing advances

- Uplink channel measurements

The Base Station Controller (BSC)

- The BSC manages the radio resources for one or more BTSs. It handles radio channel setup, frequency hopping, and handovers.
- The BSC is the connection between the mobile and the MSC. The BSC also translates the 13 Kbps voice channel used over the radio link to the standard 64 Kbps channel used by the Public Switched Telephone Network (PSDN)

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- It assigns and releases frequencies and time slots for the MS. The BSC also handles intercell handover.
 - It controls the power transmission of the BSS and MS in its area.
 - The function of the BSC is to allocate the necessary time slots between the BTS and the MSC. It is a switching device that handles the radio resources.

GSM - The Network Switching Subsystem (NSS)

- The Network switching system (NSS), the main part of which is the Mobile Switching Center (MSC), performs the switching of calls between the mobile and other fixed or mobile network users, as well as the management of mobile services such as authentication.

GSM - The Operation Support Subsystem (OSS)

- The operations and maintenance center (OMC) is connected to all equipment in the switching system and to the BSC.

- The implementation of OMC is called the operation and support system (OSS)

- Here are some of the OMC functions–

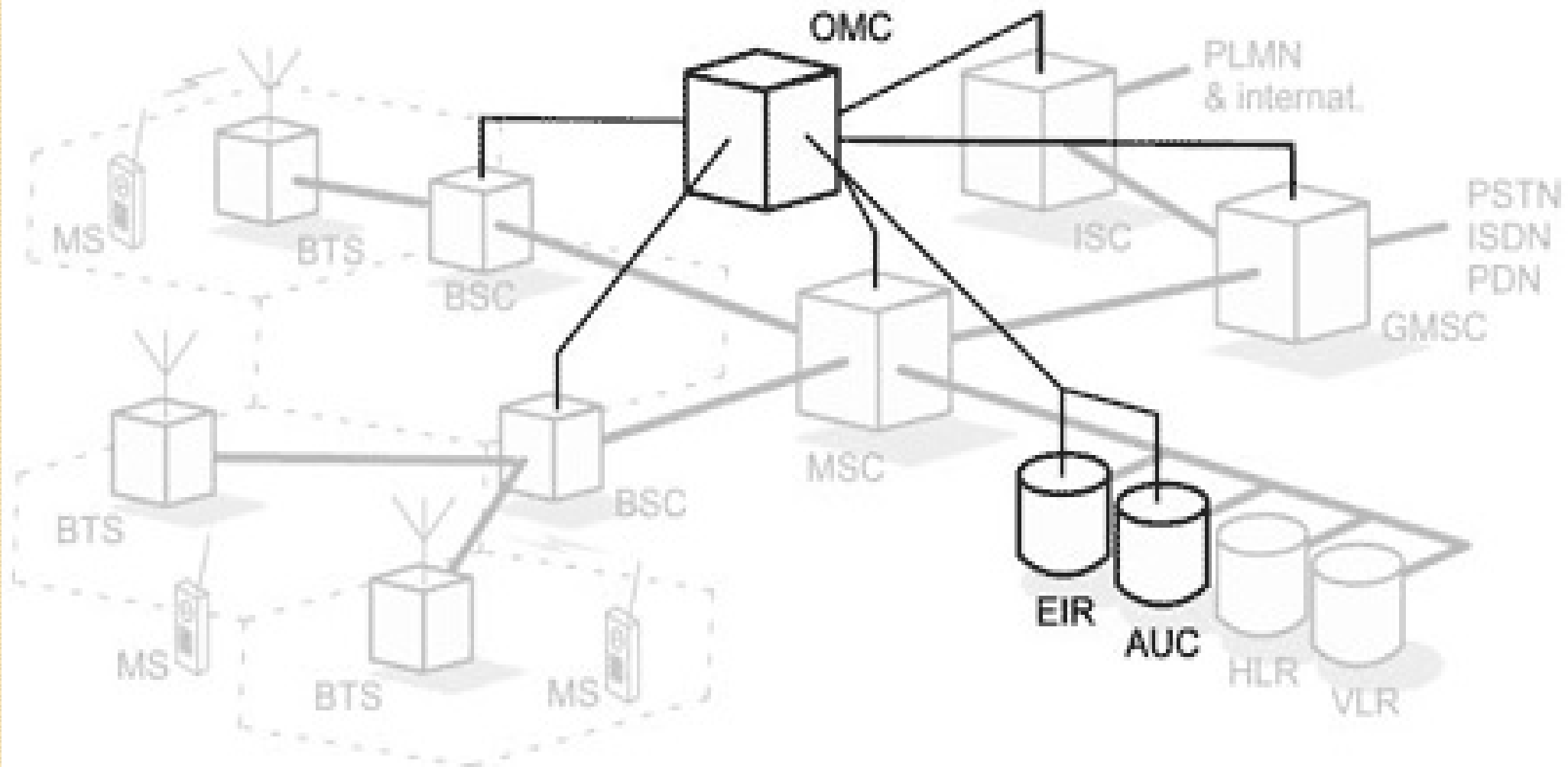
- Administration and commercial operation (subscription, end terminals, charging, and statistics).

- Security Management.

- Network configuration, Operation, and Performance Management.

- Maintenance Tasks.

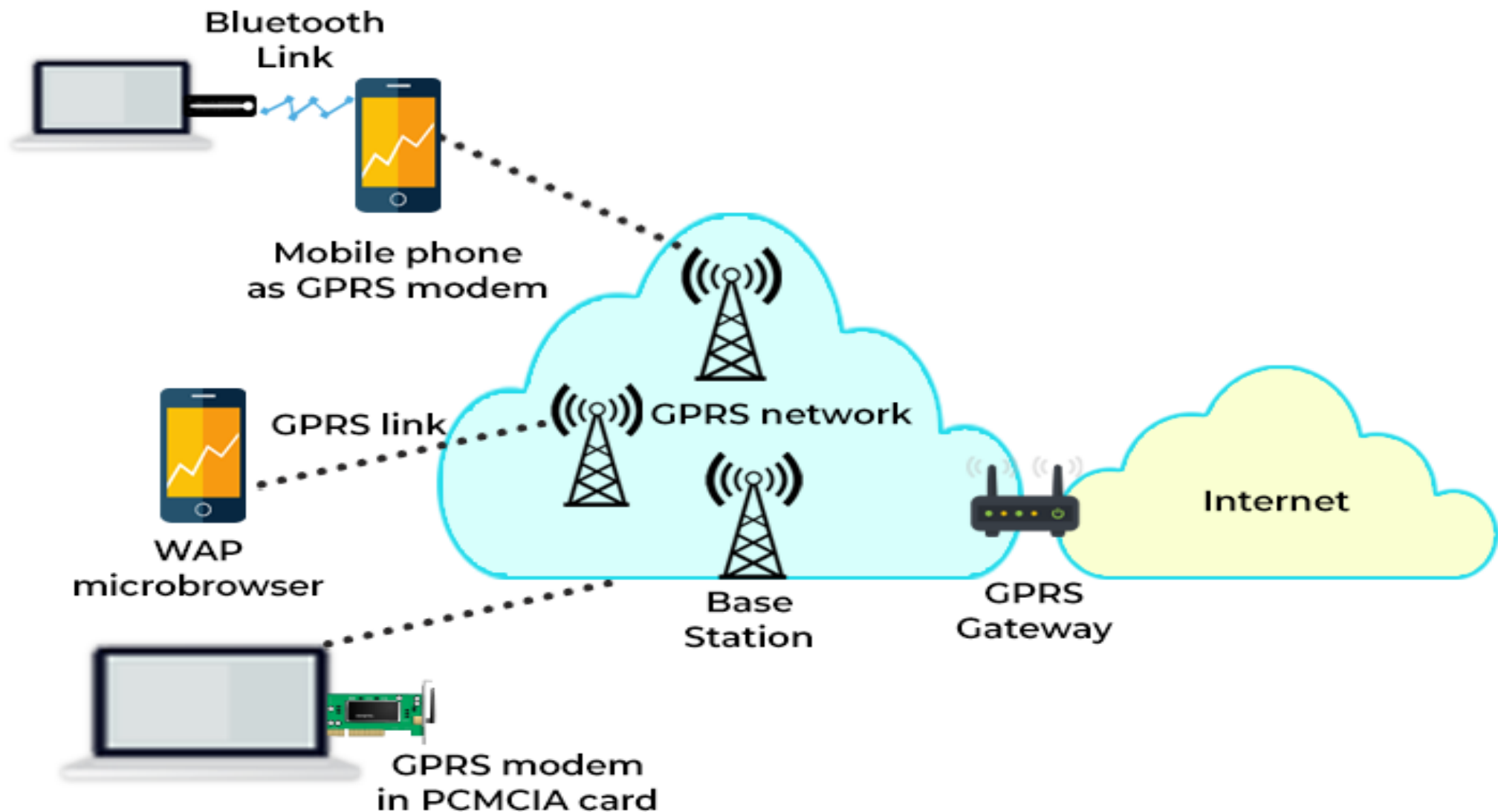
- Following is the figure, which shows how OMC system covers all the GSM



What Is GPRS?

- **General packet radio service (GPRS) is defined as a mobile communications standard that operates on 2G and 3G cellular networks to enable moderately high-speed data transfers using packet-based technologies.**

HOW DOES GPRS WORK?



How GPRS Establishes a Connected Mobile Environment

- General packet radio service (GPRS) is essentially a packet-switching technology that allows information to be transmitted via mobile networks.
- This is utilized for internet connectivity, multimedia messaging service, and other types of data transmission. GPRS is supported by GPRS cellphones, as well as laptops and handheld devices equipped with GPRS modems. Subscribers have reported downstream bandwidths of up to 80 Kbps.

Advantages and Challenges of GPRS

- Compared to the original GSM system, GPRS technology provides various perks and advantages to users and network providers.
- Companies commonly used this to create precise data capabilities using mobile communications technology.
- Let's take a look at some of the benefits of GPRS.
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BENEFITS OF GPRS

Billing made
simple



Increased
speed



“Always-on”
connectivity



Billing made simple

- When compared to circuit-switched services, GPRS packet transfer offers better consumer billing. When it comes to circuit-switched services, the cost is determined by the length of the connection.
- One of the significant disadvantages of circuit switching for IoT is cost-efficiency. The connection length determines the subscriber's operating costs in the circuit switch.

Increased speed

- A significant advantage of the GPRS system is that it provides a higher data rate than GSM.
- GSM was designed to transmit data at approximately 14.4 kbit/s at first. Over the 2G network, GPRS, on the other hand, provided data speeds up to 86kbit/s.

Always-on” connectivity

- Another benefit of GPRS is that it is always available. A GPRS connection can provide constant Internet connectivity, quick messaging, and improved SMS transfers.
- It was also the fastest network ever seen when it was first launched. For example, with a standard Global System for Mobile Communications technology, somebody could transmit only 10 SMSs per minute.

2.5G

- Enhancements that provide packet data capabilities over 2G networks. 2.5G improves the available data rates supported by the air interface, thereby permitting the introduction of new, data-oriented services and applications.
- The increased data rates rise to a theoretical maximum of 384 Kbps, although in the field available data rates may often be as low as 20 Kbps. General packet radio service (GPRS) is an example of a 2.5G technology.

Third-Generation (3G) Mobile Phones

- Third generation mobile phones, or “3G Internet” mobile phones, is a set of standards for wireless mobile communication systems, that promises to deliver quality multimedia services along with high quality voice transmission.

Features

- 3G systems comply with the International Mobile Telecommunications-2000 (IMT-2000)
- specifications by the International Telecommunication Union (ITU).
- The first 3G services were available in 1998.
- It provides high speed transmission having data transfer rate more than 0.2Mbps.
- Global roaming services are available for both voice and data.
- It offers advanced multimedia access like playing music, viewing videos, television services etc.
- It provides access to all advanced Internet services, for example surfing webpages with audio and video.

Areas of Application

- Wireless voice telephony
- Fixed wireless Internet access
- Mobile Internet access
- Video calls
- Mobile TV technologies
- Video-on-demand
- Video conferencing
- Global Positioning System (GPS)
- Location-based services

Fourth Generation (4G) mobile

- Fourth Generation (4G) mobile phones provides broadband cellular network services and is successor to 3G mobile networks. It provides an all IP based cellular communications. The capabilities provided adhere to IMT-Advanced specifications as laid down by International Telecommunication Union (ITU).

Features

- It provides an all IP packet switched network for transmission of voice, data, signals and multimedia.
- It aims to provide high quality uninterrupted services to any location at any time.
- 4G networks should have peak data rates of 100Mbps for highly mobile stations like train, car etc., and 1 Gbps for low mobility stations like residence etc.
- It also lays down that 4G networks should make it possible for 1 Gbps downlink over less than 67 MHz bandwidth.
- They provide have smooth handoffs across heterogeneous network areas.

Long – Term Evolution (LTE)

- – Long – term evolution or LTE is an extension of the 3G technology. It is a standard for high-speed mobile communication, based upon GSM/EDGE and UMTS/HSPA technologies. The peak data rate for download is 100 Mbps and upload is 50 Mbps. The LTE Advanced meets the specifications of IMT-Advanced standard for 4G technology. Its peak data rates are 1000 Mbps for downlink and 500 Mbps for uplink.

Application Areas

- Advanced mobile web access
- IP telephony
- High-resolution high-speed gaming services
- Streamed multimedia and data
- High-definition mobile TV
- Video conferencing
- 3D television

Mobile Computing Architecture

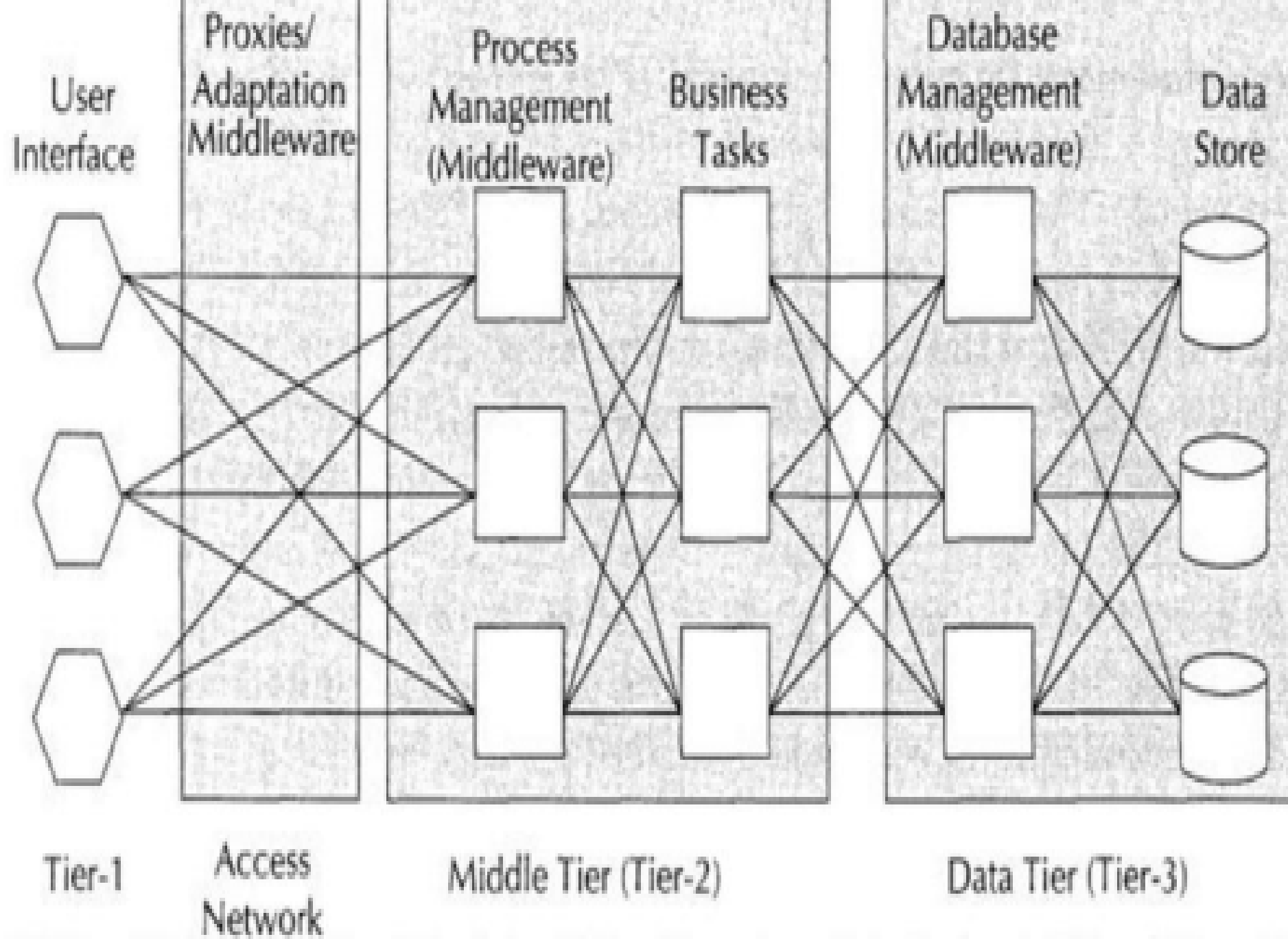
- This architecture represents the blueprint about the multiple layers in between network hardware components, devices and user application interface.
- Well – establish architecture is required for getting to access data, systematic calculations and software objects as well.

3-Tier Architecture of Mobile Computing

- Mobile computing is enabled with **three-tier architecture that has three major layers**, so here we will explore each one layer with its functions

Mobile Computing Architecture Layers:

- **Presentation Layer (UI):** This layer lets users to face device handling and rendering.
- **Application Layer (AL):** It allows executing business logic and rules.
- **Data Access Layer (DM):** It lets to get access and management.



I-Tier Presentation Layer

- This presentation layer lets to execute all applications on the client devices and provide complete user interfaces.
- It has main responsibility is to present information to user.
- Users are able to grab all information via speakers, vibration, screens, etc.
- Users can sent the information with helping of input devices like pen drives, mouse, keyboard, touch screens and so on.
- This layer is enabled with WAP browsers, customized client programs, web browsers, etc.
- Presentation layer allows accomplishing via client-side data source, Dynamic HTML and data cursors.

2-Tier Application Layer

- In the application layer, business logic performs all tasks as server for client requests from workstations. It works as business rules fetch or enter data through the Data Layer.
- It is enabled with few technologies such as PHP, . Net services, JSP, Java and so on.
- Presentation and database-independent
- This layer identifies that which types of data is required and performs as client in relation to a third tier or programming that may be situated on a mainframe computer or locally.



It takes decision on rendering, network management, security, data store access, need for many types of middlewares.

Their components are not linked to certain client, so they can be implemented by all applications and can be proceed to other locations, as responding time frame and other needed rules.

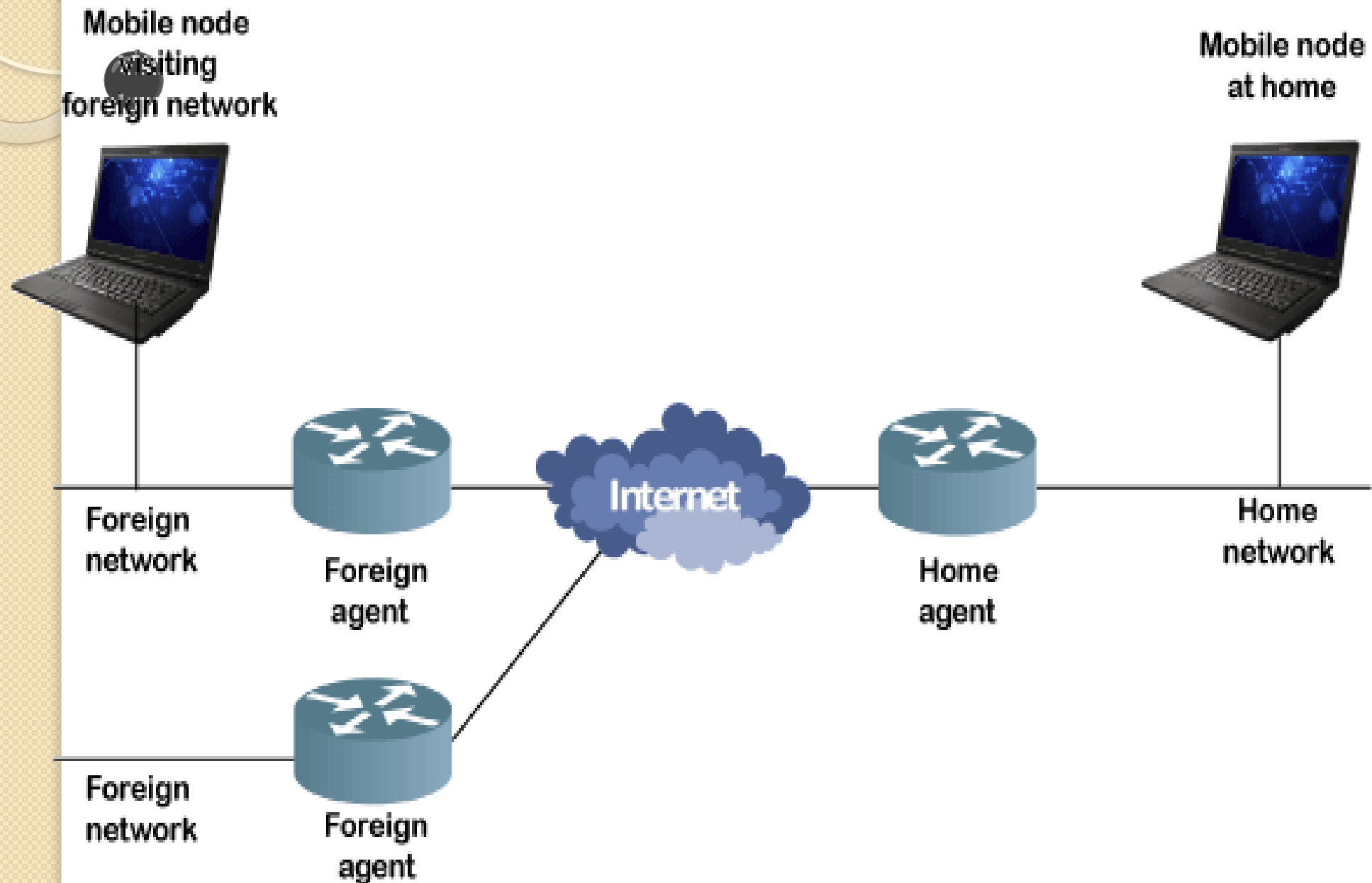
3-Tier Data Access Layer

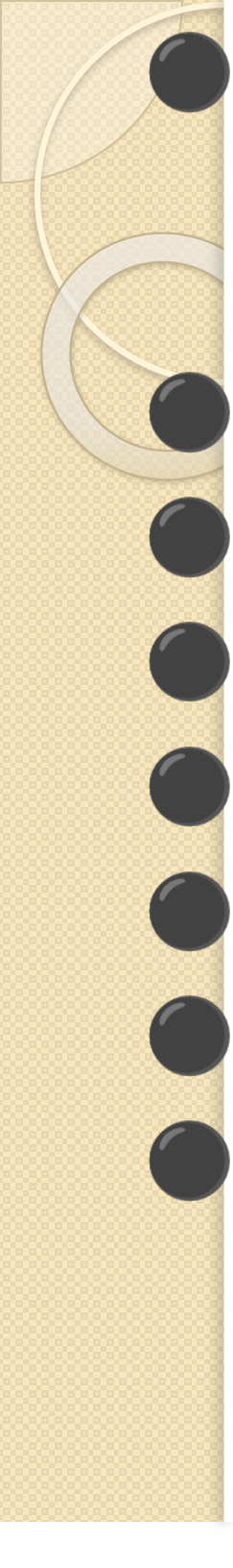
- This layer is implemented to keep store data that is required by application and work as repository for both temporary and permanent data.
- Data access layer is built up of DBMS that offers all data for above two layers.
- This layer is also known as 'DBMS Access Layer'
- All data is stored into many format like as text files or relational DB.

What is Mobile IP?

- Mobile IP or MIP is an Internet Engineering Task Force (IETF) RFC 2002, De-Facto standard communication protocol. It is created by extending Internet Protocol, IP.
- The Mobile IP allows mobile device users to move from one network to another while maintaining the same permanent IP address.
- The concept and role of Mobile IP are very important in the field of mobile computing technology.
- The mobile IP makes the communication flawless and ensures that the communication will occur without the user's sessions or connections being dropped.

The architecture of Mobile IP Technology





This is the architecture of Mobile IP technology. It consists of the following components:

- Mobile Node (MN)

- Home Agent (HA)

- Foreign Agent (FA)

- Home Network (HN)

- Foreign Network (FN)

- Corresponding Node (CN)

- Care of Address (COA)

Mobile Node

The Mobile Node is a device or a user or a router that can frequently change their network positions without changing its original IP address. Examples of mobile nodes are cell phone, personal digital assistant (PDA), laptop, etc. whose software enables network roaming capabilities.

Home Agent

The Home Agent is a router on the home network. It serves as the anchor point for communication with the Mobile Node.

Foreign Agent

The Foreign Agent is a router that provides several services such as tunneling data-grams whenever a mobile node visits a foreign network. It is responsible for delivering packets from the Home Agent to the Mobile Node.

● **Home Network**

● The home network is the base station network to which the mobile node originally belongs to.

● **Foreign Network**

● Any network other than the home network or the networks on which mobile nodes have a registered IP is called a foreign network.

● **Corresponding Node**

The partner nodes which are used for communication with mobile nodes are called corresponding nodes.

● **Care of Address**

● The Care of Address or COA is used to define the mobile node's current position or user. It is used to deliver data packets through the process of tunneling.

Working of Mobile IP

- The working of Mobile IP can be described in 3 phases:

- **Agent Discovery**

In the Agent Discovery phase, the mobile nodes discover their Foreign and Home Agents. The Home Agent and Foreign Agent advertise their services on the network using the ICMP Router Discovery Protocol (IRDP).

- **Registration**

The registration phase is responsible for informing the current location of the home agent and foreign agent for the correct forwarding of packets.

Tunneling:

This phase is used to establish a virtual connection as a pipe for moving the data packets between a tunnel entry and a tunnel endpoint.

Applications of Mobile IP

- The mobile IP technology is used in many applications where the sudden changes in network connectivity and IP address can cause problems.
- It is used in many wired and wireless environments where users have to carry their mobile devices across multiple LAN subnets.

Cellular Network

- **Cellular Network** is formed of some cells, **cell** covers a geographical region, has a base station analogous to 802.11 AP which helps mobile users attach to network and there is an air-interface of physical and link layer protocol between mobile and base station.
- All these base stations are connected to *Mobile Switching Center* which connects cells to wide area net, manages call setup and handles mobility.

Cellular Hierarchy

- **Femtocells:**
Smallest unit of the hierarchy, these cells need to cover only a few meters where all devices are in the physical range of the users.
- **Picocells:**
Size of these networks is in the range of few tens of meters, e.g., WLANs.
- **Microcells:**
Cover a range of hundreds of meters e.g. in urban areas to support PCS which is another kind of mobile technology.
- **Macro cells:**
Cover areas in the order of several kilometers, e.g., cover metropolitan areas.
- **Mega cells:**
Cover nationwide areas with ranges of hundreds of kilometers, e.g., used with satellites.

ad hoc network

- An ad hoc network is one that is spontaneously formed when devices connect and communicate with each other. The term ad hoc is a Latin word that literally means "for this," implying improvised or impromptu.
- Ad hoc networks are mostly wireless local area networks (LANs). The devices communicate with each other directly instead of relying on a base station or access points as in wireless LANs for data transfer co-ordination.
- Each device participates in routing activity, by determining the route using the routing algorithm and forwarding data to other devices via this route.

Classifications of Ad Hoc Networks

- Ad hoc networks can be classified into several types depending upon the nature of their applications.
- The most prominent ad hoc networks that are commonly incorporated are illustrated in the diagram below –

Mobile ad hoc networks (MANETs)

- This is a self-configuring, self-organising, wireless network of mobile devices.

Vehicular ad hoc networks (VANETs)

- This is network formed by communication between moving vehicles and other roadside devices.

Wireless mesh networks

- The devices connected to these networks forms a wireless mesh, depending upon the mobility patterns, nature of devices and inter-device distances.

Smart phone Ad Hoc Networks (SPANs)

- These are peer - to - peer networks created by smartphones within range of each other without requiring any cellular carrier networks, wireless access points etc.

Wireless Sensor Networks (WSN)

- Sensors are portable devices that capture specific information from environment like temperature, humidity, traffic volume etc. WSNs form ad hoc networks to capture information on the fly.

Mobile operating systems

- The operating systems found on smartphones include Symbian OS, iPhone OS, RIM's BlackBerry, Windows Mobile, Palm WebOS, Android, and Maemo. Android, WebOS, and Maemo are all derived from Linux. The iPhone OS originated from BSD and NeXTSTEP, which are related to Unix.

Popular platforms of the Mobile OS

1. **Android OS:** The Android operating system is the most popular operating system today. It is a mobile OS based on the **Linux Kernel** and **open-source software**.
2. The android operating system was developed by **Google**. The first Android device was launched in **2008**.



- **Bada (Samsung Electronics):** Bada is a Samsung mobile operating system that was launched in 2010. The Samsung wave was the first mobile to use the bada operating system. The bada operating system offers many mobile features, such as 3-D graphics, application installation, and multipoint-touch.

- **BlackBerry OS:** The BlackBerry operating system is a mobile operating system developed by **Research In Motion (RIM)**. This operating system was designed specifically for BlackBerry handheld devices. This operating system is beneficial for the corporate users because it provides synchronization with Microsoft Exchange, Novell GroupWise email, Lotus Domino, and other business software when used with the BlackBerry Enterprise Server.

- **iPhone OS / iOS:** The iOS was developed by the Apple inc for the use on its device. The iOS operating system is the most popular operating system today. It is a very secure operating system. The iOS operating system is not available for any other mobiles.
- **Symbian OS:** Symbian operating system is a mobile operating system that provides a high-level of integration with communication. The Symbian operating system is based on the java language. It combines middleware of wireless communications and personal information management (PIM) functionality. The Symbian operating system was developed by **Symbian Ltd** in **1998** for the use of mobile phones.
Nokia was the first company to release Symbian OS on its mobile phone at that time.

- **Windows Mobile OS:** The window mobile OS is a mobile operating system that was developed by **Microsoft**. It was designed for the pocket PCs and smart mobiles.
- **Harmony OS:** The harmony operating system is the latest mobile operating system that was developed by Huawei for the use of its devices. It is designed primarily for IoT devices.
- **Palm OS:** The palm operating system is a mobile operating system that was developed by **Palm Ltd** for use on personal digital assistants (PADs). It was introduced in **1996**. Palm OS is also known as the **Garnet OS**.
- **WebOS (Palm/HP):** The WebOS is a mobile operating system that was developed by **Palm**. It based on the **Linux Kernel**. The HP uses this operating system in its mobile and touchpads.

WHAT IS A MOBILE CLIENT?

- A mobile client is a technology designed to extend the features of a company's mobile telephone system. The technology makes it easier for employees to:
- Separate personal and business communications while using one smartphone
- Manage business calls while out of the office
- Transition smoothly from using a company desktop computer to a mobile device
- A mobile client is designed to make it easier to integrate portable devices into a company's communication workflows.

Benefits & Advantages of a Mobile Client

- A mobile client offers the following benefits:
- **Improved customization:** Using a mobile client, your employees can perform detailed customization on how incoming calls work through a mobile device.
- These include but are not limited to selective silent mode, ring tone filtering, and switching easily between private and business phone profiles.
- **More ways to communicate:** A mobile client makes it easier to bridge the divide between desktop and mobile communication.
- For example, a mobile client allows users to take voice messages from their computer, or receive mobile calls as though they were inside the corporate network.

What is a smart sensor?

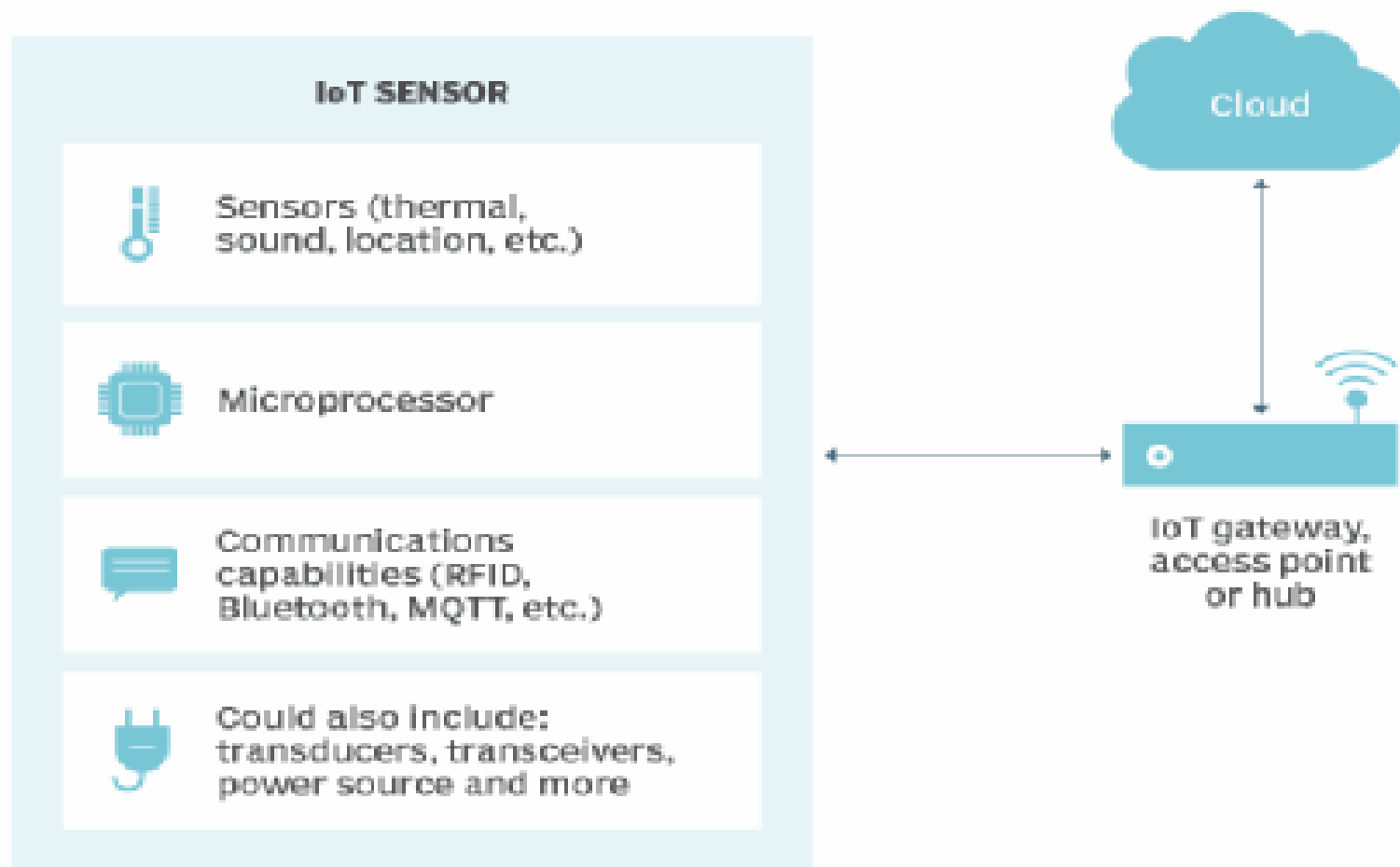
- A smart sensor is a device that takes input from the physical environment and uses built-in compute resources to perform predefined functions upon detection of specific input and then process data before passing it on.

Smart sensors enable more accurate and automated collection of environmental data with less erroneous noise amongst the accurately recorded information.

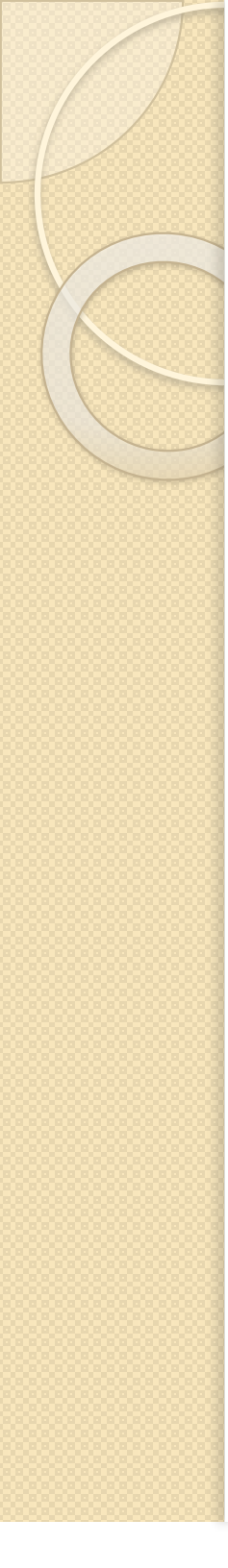
- These devices are used for monitoring and control mechanisms in a wide variety of environments including smart grids, battlefield reconnaissance, exploration and many science applications.

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- The smart sensor is also a crucial and integral element in the internet of things (IoT).
 - One implementation of smart sensors is as components of a wireless sensor and actuator network (WSAN) whose nodes can number in the thousands, each of which is connected with one or more other sensors and sensor hubs, as well as individual actuators.

An IoT sensor in action

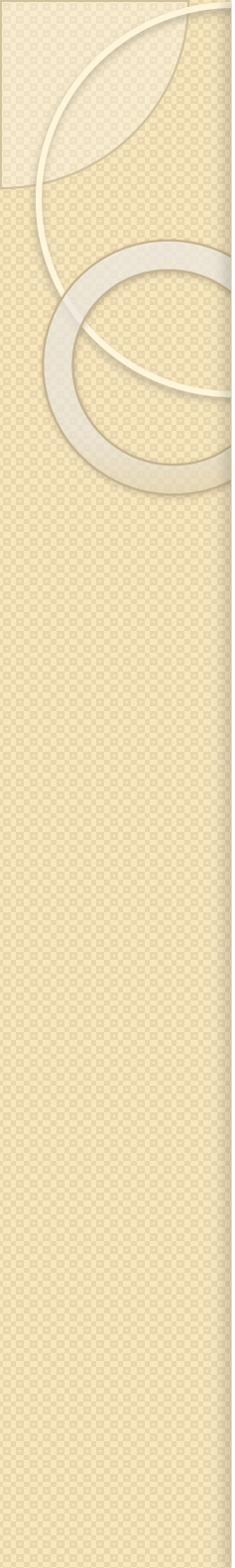


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- A smart sensor has three components: a sensor that captures data, a microprocessor that computes on the output of the sensor via programming and communications capabilities.
 - A smart sensor might also include several other components besides the primary sensor. These components can include transducers, amplifiers, excitation control, analog filters and compensation.
 - A smart sensor also incorporates software-defined elements that provide functions such as data conversion, digital processing and communication to external devices.

- 
- Smart sensors also contain integrated communications capabilities that enable them to be connected to a private network or to the internet. This enables communication to external devices.

What are different types of smart sensors?

- **Level sensors.** A level sensor is used to measure the volume of space taken up in a container. A vehicle's fuel gauge might be connected to a level sensor that monitors the level of fuel in the tank.
- **Temperature sensors.** A temperature sensor is a sensor that can monitor a component's temperature so a corrective action can be taken if necessary. In an industrial setting for example, a temperature sensor can be used to make sure machinery is not overheating.
- **Pressure sensor.** Pressure sensors are often used to monitor the pressure of gasses or fluids in a pipeline. A sudden drop in pressure might indicate a leak or a flow control issue.

- 
- **Infrared sensors.** Some infrared sensors, such as those used in thermal imaging cameras or noncontact infrared thermometers are used for temperature monitoring. Other infrared sensors are optical sensors tuned to a frequency that enables them to see light in the infrared spectrum. These types of sensors are used in medical equipment, such as pulse oximetry devices, and in electronic devices designed to be operated by remote control.
 - **Proximity sensors.** A proximity sensor is used to detect the location of a person or object with relation to the sensor. In retail environments, proximity sensors can track customer movements throughout the store.

Sensors and Actuators

Sensors and actuators allow robots to achieve certain tasks.

Examples of what a robot may need to determine include:

- How far away and where is an object (like another robot, a light, a wall)?
- At what speed is the robot moving?
- If the robot is on a hill, at what angle is it?
- If the robot is rotating, by how much has it rotated?
- Where is the robot, or in what direction is it going?
- How far it has travelled (which is odometry)?
- How much power has it consumed?
- What information is being communicated to the robot?

- 
- The robot may require internal and external sensors.

External sensors are used to measure other objects, for instance:

- How far away is an object?
- Where is the object?

Internal sensors are used to measure the robot itself, for instance:

- How fast it is going
- The angle of the robot
- Where it's 'gripper' is
- How much power it's consuming

Actuators

An actuator is a device which causes something to happen.

- This could be a robot movement, which is often achieved using motors:
- An actuator is needed to make the robots wheels turn.
- Or the joints of a robot arm to rotate.
- Or for a robot gripper to open or close.

- 
- Or it could be to let the others know what the robot is doing.
 - A light being turned on to indicate the robot is working.
 - Sound is being emitted – so the actuator is a loudspeaker.
 - Communication in some form – so a transmitter is needed.
 - An LCD (Liquid Crystal Display) screen for showing pictures and other data.

SMART HOME APPLIANCE

- Faststream Technologies Smart home appliance solutions are used in a smart house and work on smart home automation technology that uses electric power transmission where the signals have short radio frequency which uses open source software, proprietary software and distributed denial-of-service attack (DDoS).
-

Smart Lawn Mower

A high-quality lawn mower needs rigorous motor control and sensing, high efficiency, secure connectivity, and high reliability and robustness. Using a safe wireless connection to the smartphone, lawnmowers are used easily prolonged to make its best use of their lifespan and presentation.

Smart Refrigerator

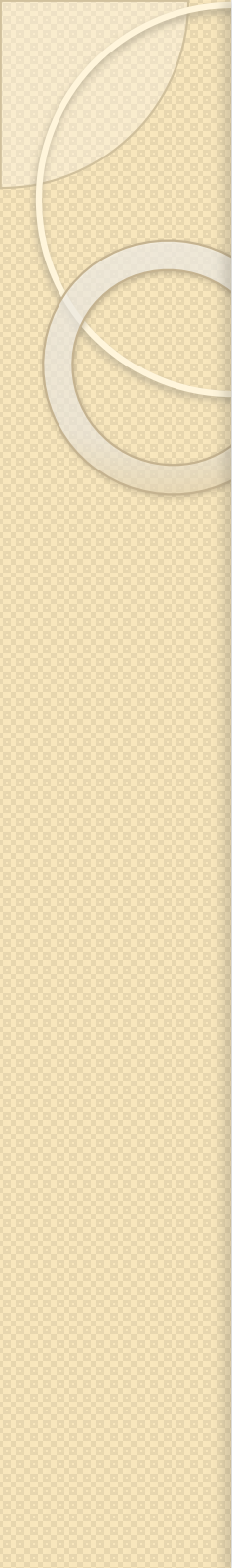
A normal quality-drive refrigerator is used for high reliability and robustness, high efficiency, safe and accurate sensing. This entitles the expansion of a refrigerator with the least defrost cycles, stable wireless connectivity, and higher user occurrence work on smart home automation technology.

PLMN(Public Land Mobile Network)

- The PLMN is the term used to describe all mobile wireless networks that use earth-based stations rather than satellites.
- PLMN is the mobile equivalent of the PSTN. Such a system can stand alone, but often it is interconnected with a fixed system such as the public switched telephone network (PSTN).
- The most familiar example of a PLMN end user is a person with a cell phone. However, mobile and portable Internet use is also common.

PLMN code

- A Public Land Mobile Network is identified by a globally unique PLMN code, which consists of a MCC (Mobile Country Code) and MNC (Mobile Network Code).
- Hence, it is a five- to six-digit number identifying a country, and a mobile network operator in that country, usually represented in the form 001-01 or 001-001.

- 
- A PLMN is part of a:
 - Location Area Identity (LAI) (Public Land Mobile Network and Location Area Code)
 - Cell Global Identity (CGI) (LAI and Cell Identifier)
 - IMSI (see PLMN code and IMSI)

PLMN services

- A Public Land Mobile Network typically offers the following services to a mobile subscriber:
- Emergency calls to local Fire/Ambulance/Police stations.
- Voice calls to/from any other PLMN (“cellular network”) or PSTN (“landline”/VoIP).
- Short Messaging Service (SMS) services to/from any other PLMN or SIP service (the original form of texting on a mobile phone, now often replaced by Messaging apps).
- Multimedia Messaging Service (MMS) services to/from any other PLMN or SIP service.
- Unstructured Supplementary Service Data (USSD) for operator specific interactions (e.g. dialing “*#100#” to indicate the current balance).
- Internet data connectivity for arbitrary services, e.g. via GPRS in GSM, LuPS in UMTS, or LTE.

High Speed Circuit Switched Data (HSCSD)

- It is an enhancement in the data rate of circuit switched data in a GSM network. HSCSD uses two techniques to increase the data rate.
- First, HSCSD makes it possible to use more than one time slot. GSM uses time division multiple access (TDMA).
- Each radio channel is divided in eight time slots. Each time slot is allocated to a different user.
- This makes it possible to serve eight customers on one radio channel. HSCSD makes it possible to allocate more than one time slot to a user.

- 
- The second technique used by HSCSD is that the error correction can be adapted to the quality of the radio channel.
 - A standard slot can carry 9.6 kbps. HSCSD makes it possible to increase this to 14.4 kbps. The quality of the radio channel must be good enough to do so.
 - The maximum data rate of a HSCSD configuration with 14.4-kbps channel coding is 115.2 kbps, if all eight time slots are allocated to the same user.

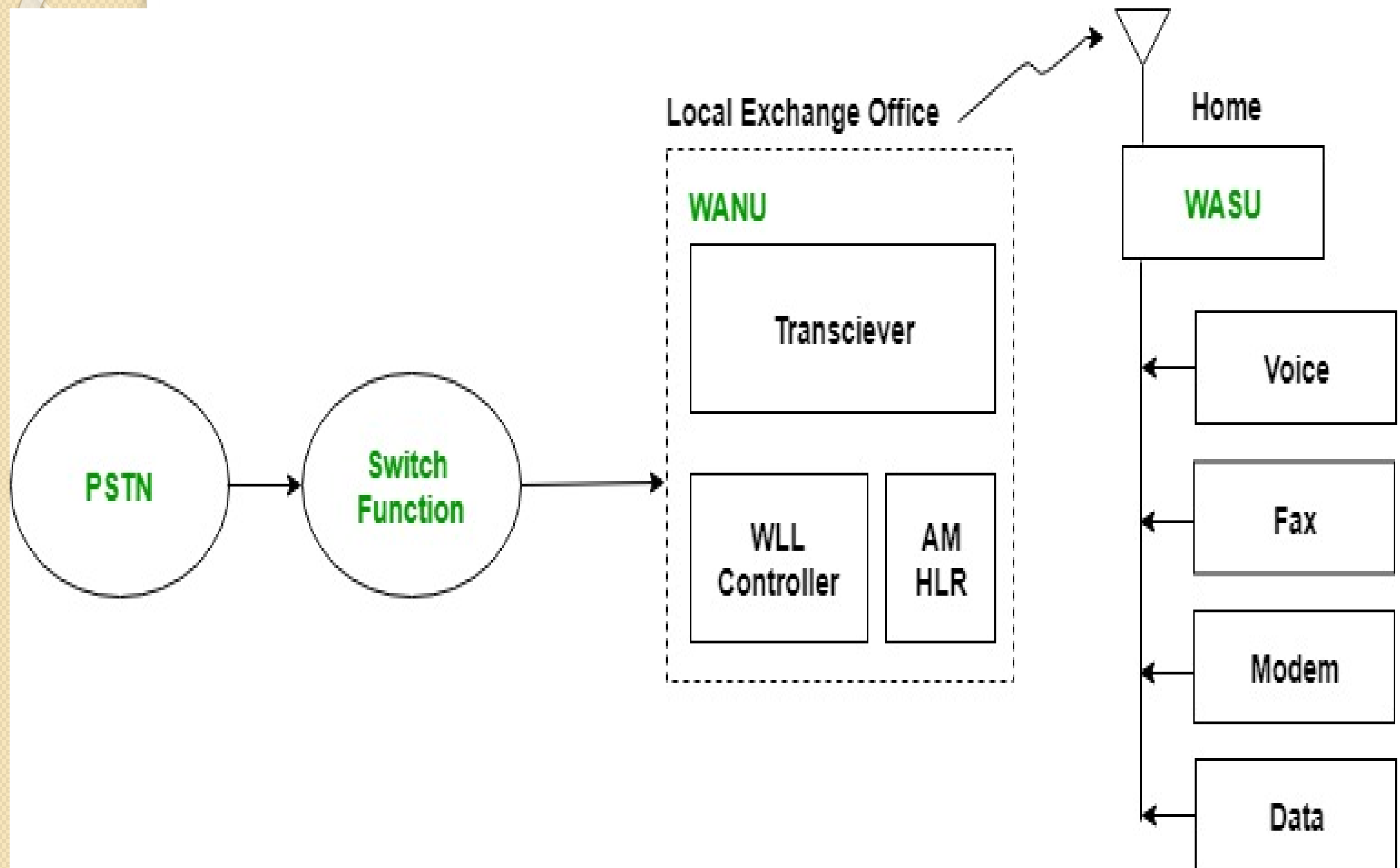


The main **benefit of HSCSD** compared to other data enhancements introduced later is that it is an inexpensive way to implement higher data rates in GSM networks. There modifications to be made are relatively small.

wireless local loop (WLL)

- Local loop is a circuit line from a subscriber's phone to the local central office (LCO).
- But the implementation of local loop of wires is risky for the operators, especially in rural and remote areas due to less number of users and increased cost of installation.
- Hence, the solution for it is the usage of wireless local loop (WLL) which uses wireless links rather than copper wires to connect subscribers to the local central office.

Architecture of WLL



WLL Component

PSTN:

It is Public Switched Telephone Network which is a circuit switched network. It is a collection of world's interconnected circuit switched telephone networks.

Switch Function:

Switch Function switches the PSTN among various WANUs.

WANU:

It is short for Wireless Access Network Unit. It is present at the local exchange office. All local WASUs are connected to it. Its functions includes: Authentication, Operation & maintenance, Routing, Transceiving voice and data.

WASU:

It is short for Wireless Access Subscriber Units. It is present at the house of the subscriber. It connects the subscriber to WANU and the power supply for it is provided locally.

Advantages of WLL:

- It eliminates the first mile or last mile construction of the network connection.
- Low cost due to no use of conventional copper wires.
- Much more secure due to digital encryption techniques used in wireless communication.
- Highly scalable as it doesn't require the installation of more wires for scaling it.

● Features of WLL:

- Internet connection via modem
- Data service
- Voice service
- Fax service

Application of WLL

- There are two types of WLL :- 1 LMDS2 MMDS
- LMDS provides services :- High speed internet access , Real time multimedia file transfer, remote access to local area network, interactive video , video on demand , video conference , telephone services
- MMDS can transmit videos, voice or data signals at rate 1.5 to 2 Mbps downstream and 320 Kpbs to upstream

4G and 5G Networks

High-Speed Packet Access (HSPA)

High-speed packet access (HSPA) refers to a set of technologies that came about as a result of enhancements to wideband code division multiple access (WCDMA) systems.

HSPA is composed of the High-Speed Downlink Packet Access (HSDPA) protocol and the High-Speed Uplink Packet Access (HSUPA) protocol. It features peak data rates of up to 14 Mbps downlink and up to 5.7 Mbps uplink.

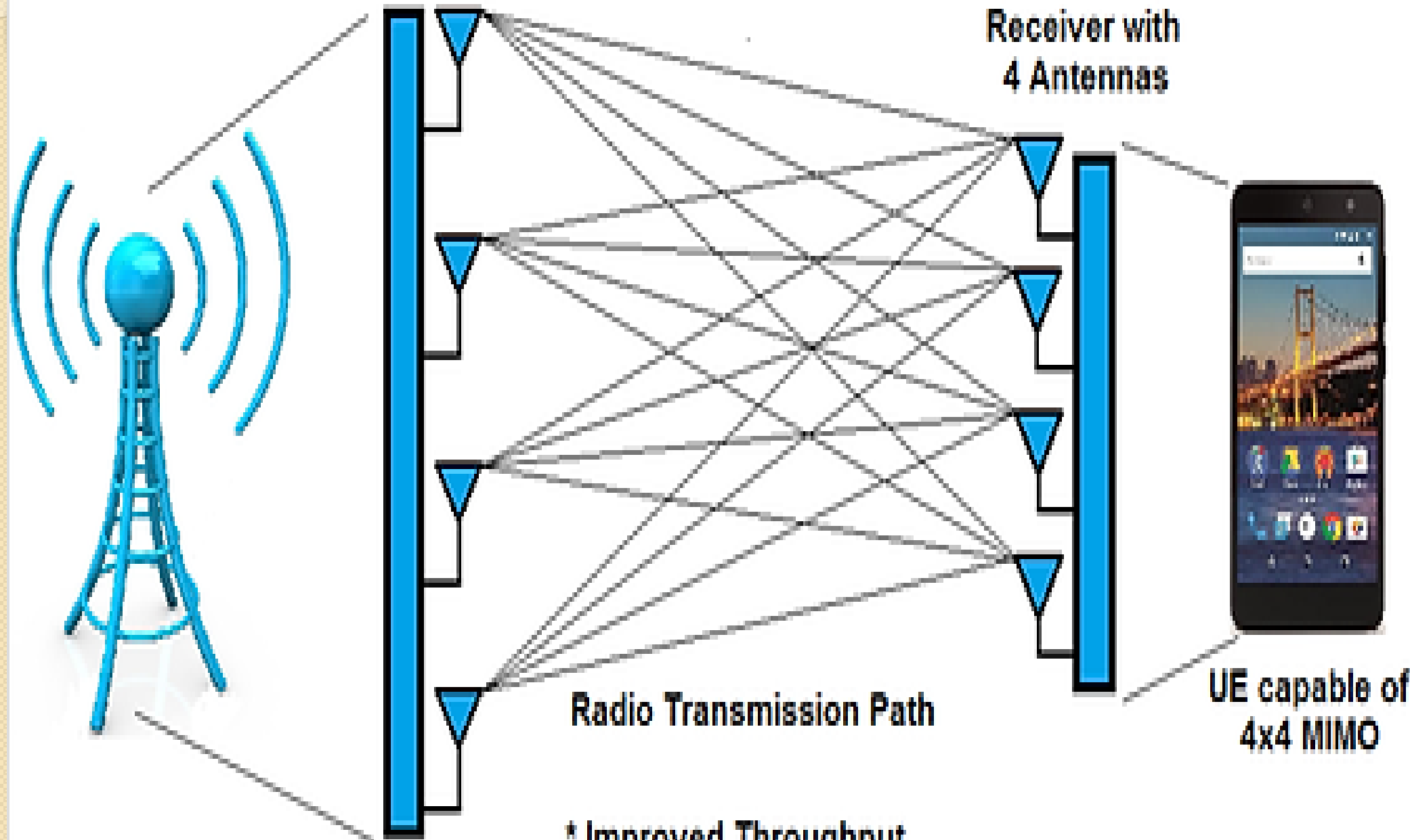
MIMO (multiple input, multiple output)

- MIMO (multiple input, multiple output) is **an antenna technology for wireless communications in which multiple antennas are used at both the source (transmitter) and the destination (receiver).**
- MIMO communication sends the same data as several signals simultaneously through multiple antennas, while still utilizing a single channel. This is a form of antenna diversity, which uses multiple antennas to improve signal quality and strength of an RF link.

- 
- This data is split into multiple data streams at the transmission point and recombined on the receive side by another MIMO radio configured with the same number of antennas.
 - The receiver is designed to take into account the slight time difference between receptions of each signal, any additional noise or interference, and even lost signals.

**LTE eNB with
4 Transmitters**

**Receiver with
4 Antennas**



**UE capable of
4x4 MIMO**

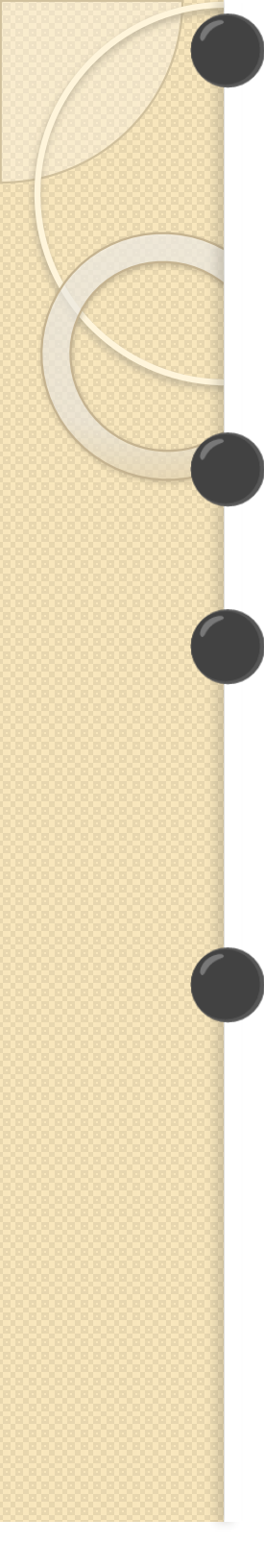
- * Improved Throughput
- * High Data Rate
- * Increased Coverage

WiMAX and LTE(Long term Evolution)

- WiMAX, the **Worldwide Interoperability for Microwave Access**, is a telecommunications technology aimed at providing wireless data over long distances in a variety of ways, from point-to-point links to full mobile cellular type access.
- WiMAX and LTE are both wireless network technologies suitable for data connections to mobile (and sometimes stationary) devices.
- WiMAX is an IEEE standard, 802.16; its original name is WirelessMAN (for Metropolitan Area Network), and this name appears intermittently in the IEEE standards.

- 
- Long-Term Evolution (LTE) was first introduced in 2008, it defined a new cellular access network with high spectral efficiency, high peak data rates, short round trip time as well as flexibility in frequency and bandwidth.
 - It signifies an evolving level of performance as the capabilities of cellular hardware, software and network technology — such as speed, latency, battery usage and cost efficiency

- 
- LTE technology has delivered multiple benefits worldwide: LTE connectivity is almost *universally available* around the world for both consumer and commercial/industrial applications.
 - LTE provides *long-term network continuity* as older networks such as 2G and 3G sunset.
 - In regions where 5G will not be available for some time, 4G LTE, 4G LTE Advanced, and 4G LTE Advanced Pro technology will *support migration needs from 2G/3G for years to come*.
 - LTE offers *higher speeds* as well as significant benefits for low power applications and simpler, low-cost devices — providing a single technology foundation for a wide variety of use cases.

- 
- Here are a few important facts to know about how LTE works at a high level: *LTE offers lower latency* and increased throughput throughout the network, dramatically improving upon 3G network performance.
 - *LTE operates on a separate spectrum* from 3G networks and requires new hardware.
 - *LTE provides fast data download speeds* of several 100th megabits per second (Mbps), compared to several 10th Mbps for 3G, meaning that LTE is 5-10 times faster than 3G.
 - *LTE can support data, voice (VoLTE), instant messaging and video* on smartphones and tablets over a single interface. With 3G, this was done over different systems, and on some networks voice and data was mutually exclusive.

Differences between 3G and LTE:

Here are some of the major

- **Speed:** 3G is slower, with data rates measured in kilobits per second (Kbps) rather than megabits per second.

- **Latency:** 3G latency (the time gap between when data is sent and when it is received) is much greater.

- **Power usage:** LTE devices transfer a higher volume of data and can therefore use batteries faster than 3G, which has cost and power management implications that developers and network managers must take into consideration.

- **Availability and reliability:** 3G networks until recently were more widely available. Today, 4G networks are nearly universally available and the reliability differences have largely disappeared.

What is Ultra Wideband?

- UWB is a short-range, wireless communication protocol that – like Bluetooth or Wi-Fi – uses radio waves. But it differs substantially in that it operates at a very high frequency. As its name denotes, it also uses a wide spectrum of several GHz.

What is 5G?

- 5G is the 5th generation mobile network. It is a new global wireless standard after 1G, 2G, 3G, and 4G networks.
- 5G enables a new kind of network that is designed to connect virtually everyone and everything together including machines, objects, and devices.
- 5G wireless technology is meant to deliver higher multi-Gbps peak data speeds, , more reliability, massive network capacity, increased availability, and a more uniform user experience to more users.
- Higher performance and improved efficiency empower new user experiences and connects new industries.

Features of 5G

- **Enhanced mobile broadband**
 ,more uniform data rates, lower latency, and lower cost-per-bit.

Mission-critical communications

5G can enable new services that can transform industries with ultra-reliable, available, low-latency links like remote control of critical infrastructure, vehicles, and medical procedures.

Massive IoT

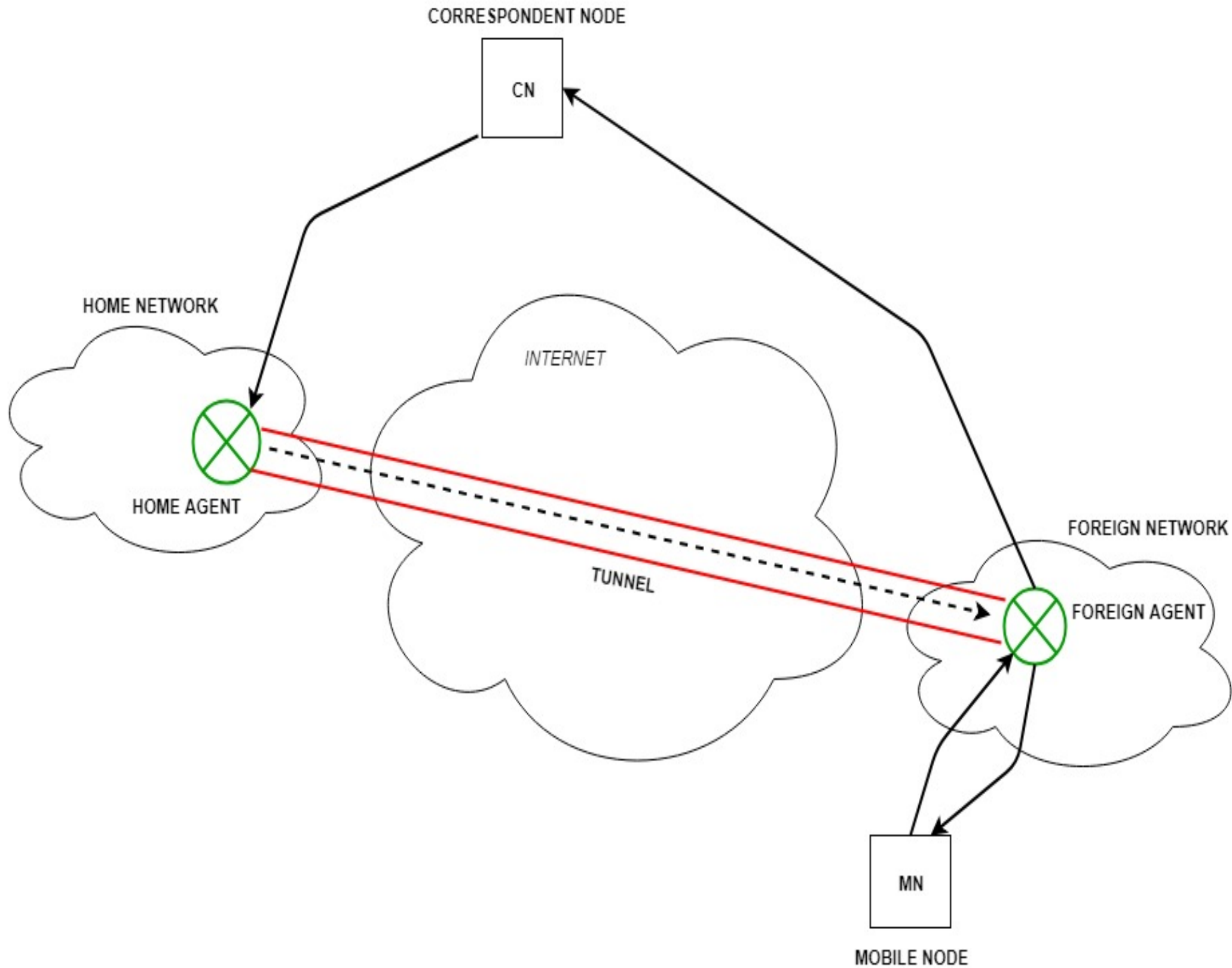
5G is meant to seamlessly connect a massive number of embedded sensors in virtually everything through the ability to scale down in data rates, power, and mobility—providing extremely lean and low-cost connectivity solutions.

Mobile IP networking layer

- **Working:**
- The correspondent node sends the data to the mobile node.
- Data packets contain the correspondent node's address (Source) and home address (Destination). Packets reach the home agent.
- But now mobile node is not in the home network, it has moved into the foreign network. The foreign agent sends the care-of-address to the home agent to which all the packets should be sent.
- Now, a tunnel will be established between the home agent and the foreign agent by the process of tunneling.

Key Mechanisms in Mobile IP:

- **Agent Discovery:** Agents advertise their presence by periodically broadcasting their agent advertisement messages. The mobile node receiving the agent advertisement messages observes whether the message is from its own home agent and determines whether it is in the home network or foreign network.
- **Agent Registration:** Mobile node after discovering the foreign agent sends a registration request (RREQ) to the foreign agent. The foreign agent, in turn, sends the registration request to the home agent with the care-of-address. The home agent sends a registration reply (RREP) to the foreign agent. Then it forwards the registration reply to the mobile node and completes the process of registration.
- **Tunneling:** It establishes a virtual pipe for the packets available between a tunnel entry and an endpoint. It is the process of sending a packet via a tunnel and it is achieved by a mechanism called encapsulation. It takes place to forward an IP datagram from the home agent to the care-of-address. Whenever the home agent receives a packet from the correspondent node, it encapsulates the packet with source address as home address and destination as care-of-address.



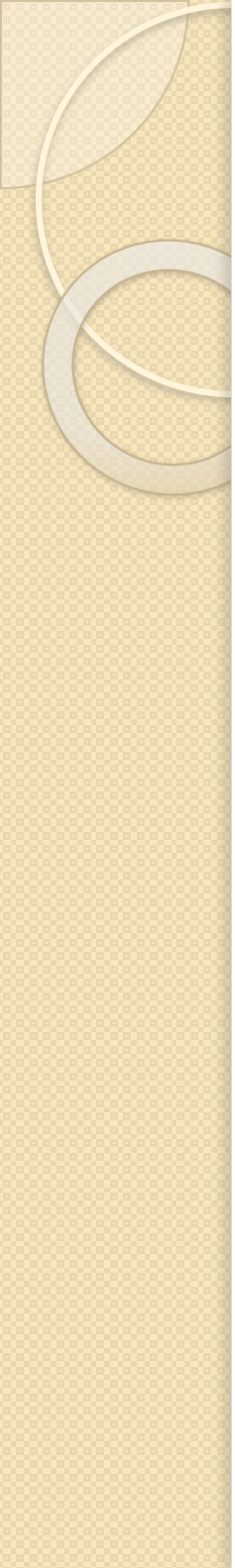
- 
- Tunneling establishes a virtual pipe for the packets available between a tunnel entry and an endpoint.
 - It is the process of sending a packet via a tunnel and it is achieved by a mechanism called **encapsulation**.
 - The home agent encapsulates the data packets into new packets in which the source address is the home address and destination is the care-of-address and sends it through the tunnel to the foreign agent.
 - Foreign agent, on another side of the tunnel, receives the data packets, decapsulates them, and sends them to the mobile node. The mobile node in response to the data packets received sends a reply in response to the foreign agent.
 - The foreign agent directly sends the reply to the correspondent node.

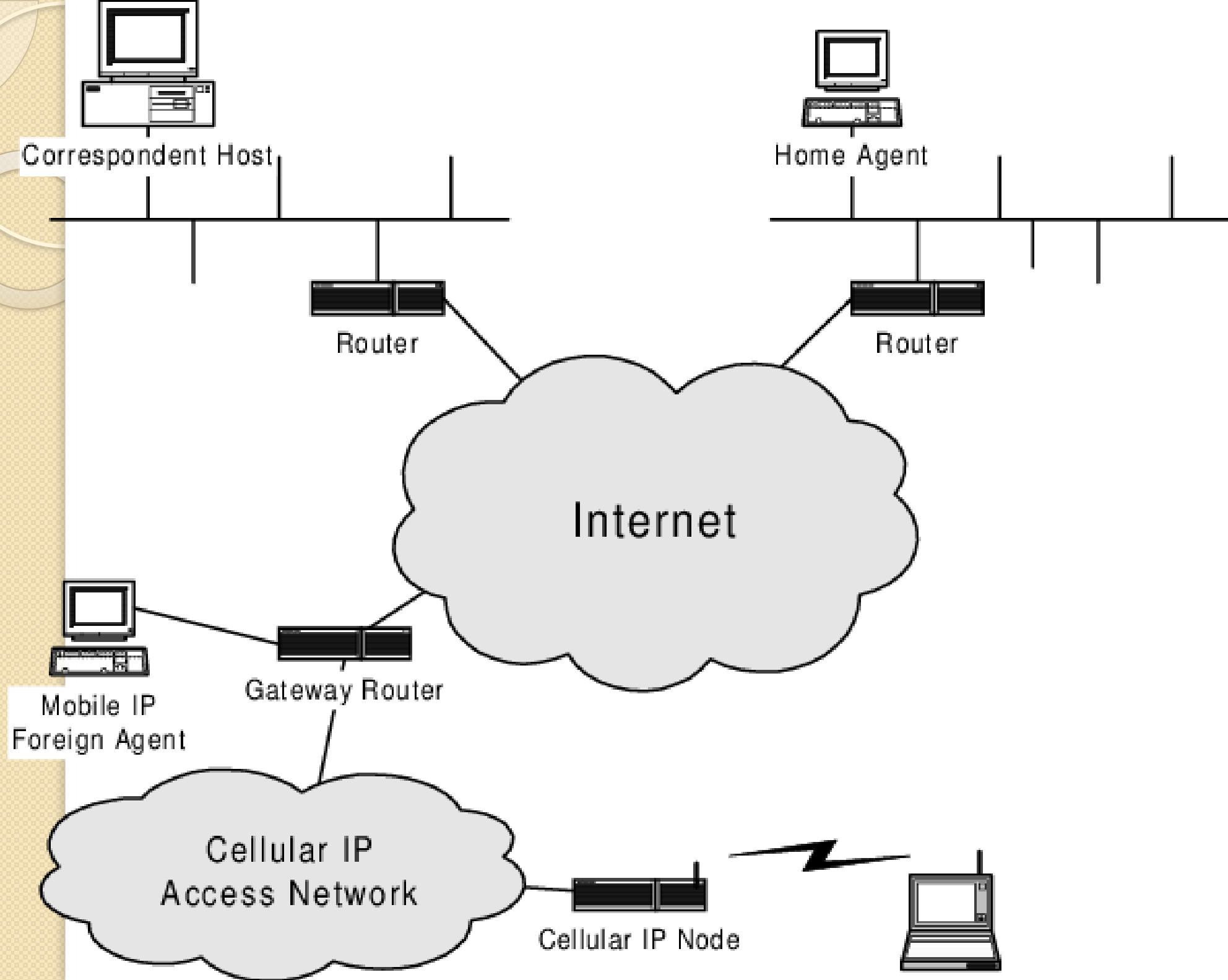
Route Optimization in Mobile IP:

- The route optimization adds a conceptual data structure, the binding cache, to the correspondent node.
- The binding cache contains bindings for the mobile node's home address and its current care-of-address.
- Every time the home agent receives an IP datagram that is destined to a mobile node currently away from the home network, it sends a binding update to the correspondent node to update the information in the correspondent node's binding cache.
- After this, the correspondent node can directly tunnel packets to the mobile node. Mobile IP is provided by the network providers.

The Cellular IP

- The Mobile IP protocol is considered to have limitations in its capability to handle large numbers of Mobile Stations moving fast between different radio cells.
- The Handover frequency should typically not exceed once a second. However, Mobile IP is well suited for interconnecting disparate cellular networks effectively providing global mobility.
- Resulting from this some micro mobility approaches have been proposed within the IETF, which support mobility in a well defined area,

- 
- Four fundamental design principles of the protocol are:
 - location information is stored in distributed data bases,
 - location information referring to a mobile host is created and updated by regular IP datagrams originated by the said mobile host
 - location information is stored as soft state
 - location management for idle mobile hosts is separated from location management of hosts that are actively transmitting or receiving data



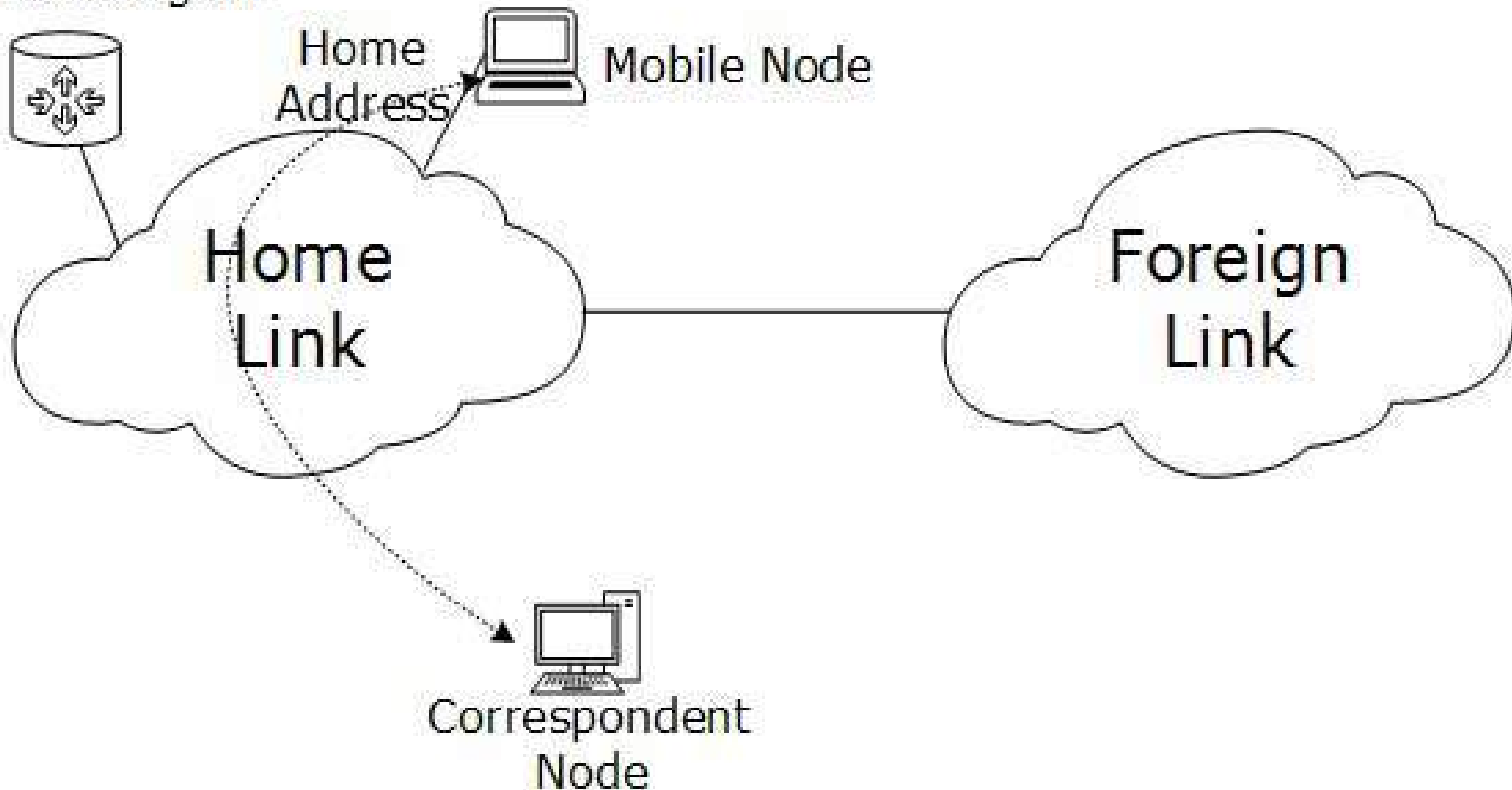
IPv6 mobility

- When a host is connected to a link or network, it acquires an IP address and all communication take place using that IP address on that link. As soon as, the same host changes its physical location, that is, moves into another area / subnet / network / link, its IP address changes accordingly, and all the communication taking place on the host using old IP address, goes down.
- IPv6 mobility provides a mechanism for the host to roam around different links without losing any communication/connection and its IP address.

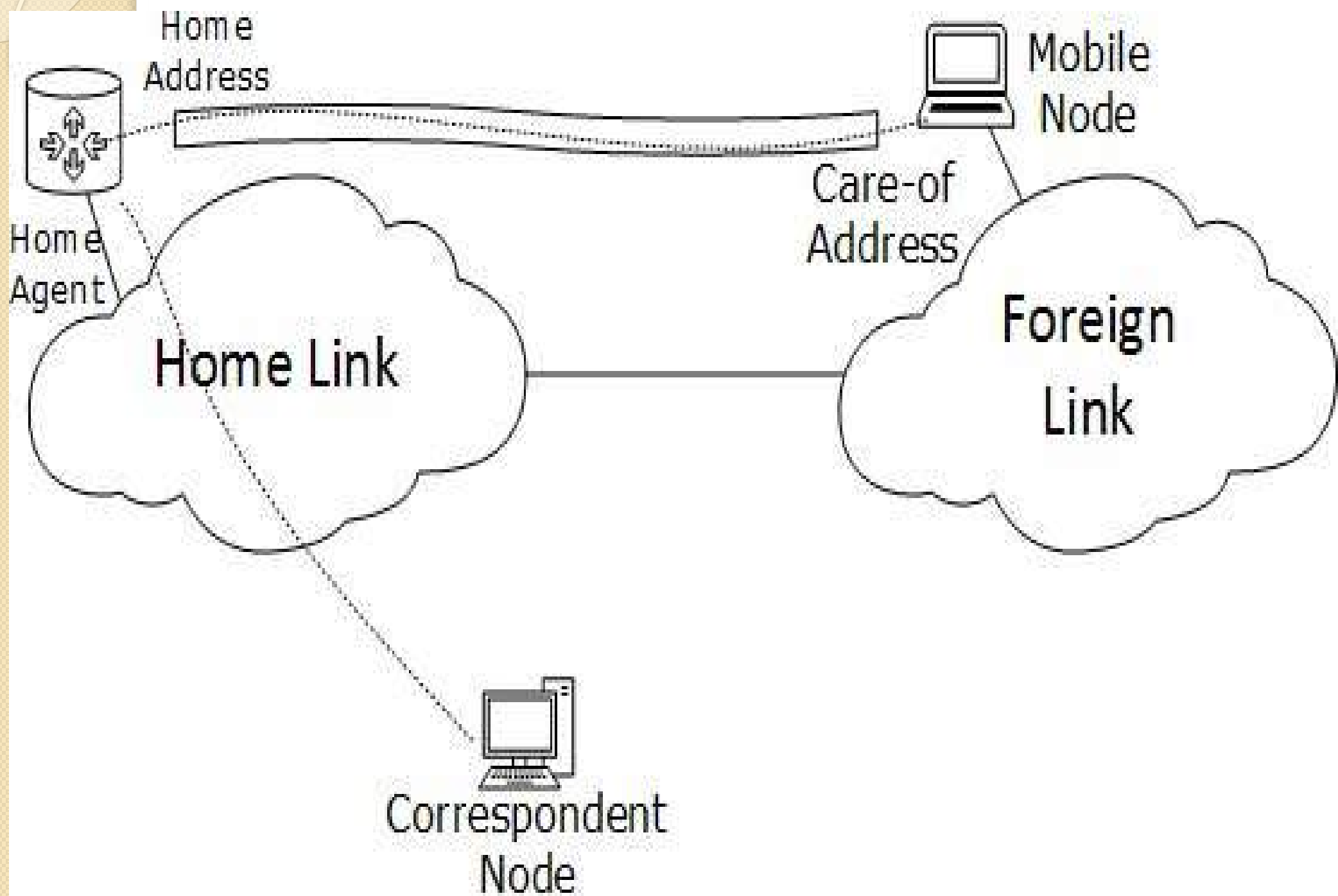
Mobility Operation

When Mobile Node stays in its Home Link, all communications take place on its Home Address as shown below:

Home Agent



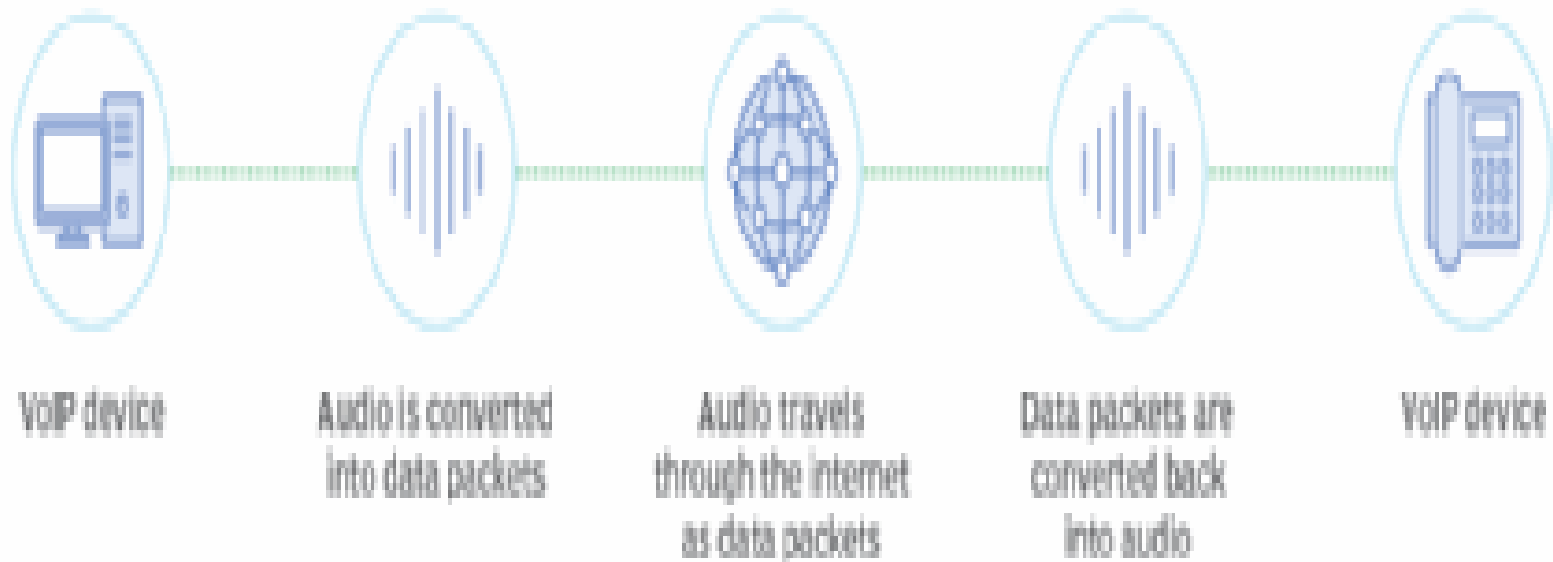
- 
- When a Mobile Node leaves its Home Link and is connected to some Foreign Link, the Mobility feature of IPv6 comes into play. After getting connected to a Foreign Link, the Mobile Node acquires an IPv6 address from the Foreign Link. This address is called Care-of Address.
 - The Mobile Node sends a binding request to its Home Agent with the new Care-of Address. The Home Agent binds the Mobile Node's Home Address with the Care-of Address, establishing a Tunnel between both.
 - Whenever a Correspondent Node tries to establish connection with the Mobile Node (on its Home Address), the Home Agent intercepts the packet and forwards to Mobile Node's Care-of Address over the Tunnel which was already established.



Voice over Internet Protocol (VoIP),

- Voice over Internet Protocol (VoIP), is a technology that allows to make voice calls using a broadband Internet connection instead of a regular (or analog) phone line
- VoIP services convert voice into a digital signal that travels over the Internet. If we are calling a regular phone number, the signal is converted to a regular telephone signal before it reaches the destination.
- VoIP can allow you to make a call directly from a computer, a special VoIP phone, or a traditional phone connected to a special adapter.

How VoIP works



Advantages of VoIP

- Some VoIP services offer features and services that are not available with a traditional phone, or are available but only for an additional fee.
- Paying for both a broadband connection and a traditional telephone line can be avoided.
- Smoother connection than an analog signal can be provided.

Disadvantages of VoIP –

- Some VoIP services don't work during power outages and the service provider may not offer backup power.
- Not all VoIP services connect directly to emergency services through emergency service numbers.
- VoIP providers may or may not offer directory assistance.

The **IP security (IPSec)**

- The **IP security (IPSec)** is an Internet Engineering Task Force (IETF) standard suite of protocols between 2 communication points across the IP network that provide data authentication, integrity, and confidentiality.
- It also defines the encrypted, decrypted and authenticated packets. The protocols needed for secure key exchange and key management are defined in it.

Uses of IP Security

IPsec can be used to do the following things:

- To encrypt application layer data.
- To provide security for routers sending routing data across the public internet.
- To provide authentication without encryption, like to authenticate that the data originates from a known sender.
- To protect network data by setting up circuits using IPsec tunneling in which all data is being sent between the two endpoints is encrypted, as with a Virtual Private Network(VPN) connection.

Working of IP Security

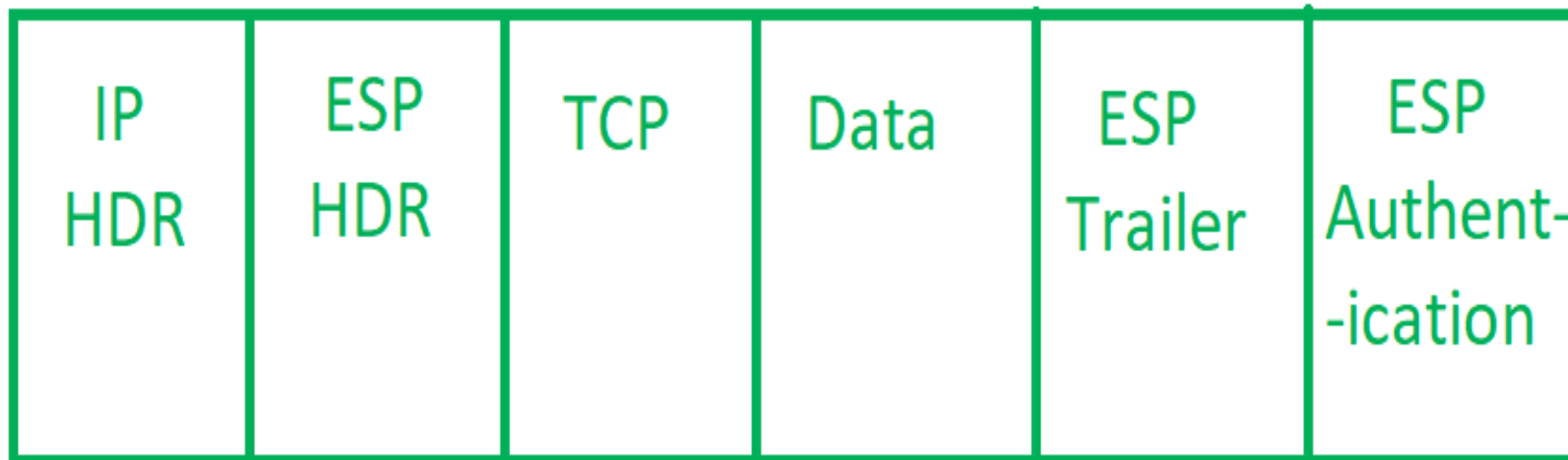
- The host checks if the packet should be transmitted using IPsec or not. These packet traffic triggers the security policy for themselves. This is done when the system sending the packet apply an appropriate encryption. The incoming packets are also checked by the host that they are encrypted properly or not.
- Then the **IKE Phase I** starts in which the 2 hosts(using IPsec) authenticate themselves to each other to start a secure channel. It has 2 modes. The **Main mode** which provides the greater security and the **Aggressive mode** which enables the host to establish an IPsec circuit more quickly.

- 
- The channel created in the last step is then used to securely negotiate the way the IP circuit will encrypt data across the IP circuit.
 - Now, the **IKE(internet key exchange) Phase 2** is conducted over the secure channel in which the two hosts negotiate the type of cryptographic algorithms to use on the session and agreeing on secret keying material to be used with those algorithms.
 - Then the data is exchanged across the newly created IPsec encrypted tunnel. These packets are encrypted and decrypted by the hosts using IPsec Sas(security association).
 - When the communication between the hosts is completed or the session times out then the IPsec tunnel is terminated by discarding the keys by both the hosts.

Esp-encapsulating security payload



Original Packet



← Encryption →

← Authentication →

Virtual Networking

- Virtual networking is a technology that facilitates the control of one or more remotely located computers or servers over the Internet. Here data communication occurs between two or more virtual machines.
- It is similar to traditional networking but interconnection occurs through virtual computing environment means in virtual networking all devices, servers, virtual machines are connected through software and wireless technology.

- 
- It is based on physical networking principle but its functions are mostly software-driven. In virtual networking devices across many locations connects with each other and function with the same capabilities as a traditional physical network.
 - Virtual networking reduces the cost and complexity of operating and maintaining hardware. Virtual networking software are installed on remote computers or servers to avail of the virtual networking facility.

Working of virtual networking :

Virtual networking interconnects remote devices and machines from any location using the software.

Some vendors offer comprehensive virtual networking software and services.

Modern technology is used to create an extended network that works wirelessly.

Two classes of Virtual Networking :

- 1. Virtual Private Network –
- A Virtual Private Network in short called as VPN. VPN uses the internet to connect two or more existing networks. It is an internet-based virtual networking technology that allows to access any physical networks that are connected. It allows connecting any private network through the internet securely and privately. It is just like a private point to point connection between two devices or networks.
- 2. Virtual Local Area Network –
- Virtual Local Area Network in short called VLAN. VLAN subdivides the LAN logically into different broadcast domains means it uses partitions to group devices on a LAN network into domains with resources and configurations. It allows better security, monitoring, and management of the devices and servers within a specific domain. Each VLANs act as a separate LAN. Data transmission becomes much easier and with VLAN the increasing transmission demand is fulfilled.

Benefits of virtual networking :

- Flexibility and scalability
- Digital security
- Low networking hardware investment
- Less overall traffic
- Centralized network management
- Easy address to issues
- Cost savings
- Remote work capabilities
- Productivity

Mobile Cloud Computing

Mobile Cloud Computing which is defined as a combination of mobile computing, cloud computing, and wireless network that come up together purpose such as rich computational resources to mobile users, network operators, as well as to cloud computing providers.

Mobile Cloud Computing is meant to make it possible for rich mobile applications to be executed on a different number of mobile devices. In this technology, data processing, and data storage happen outside of mobile devices.

Advantages:

-
- Extended battery life.
- Improvement in data storage capacity and processing power.
- Improved synchronization of data due to “store in one place, accessible from anywhere ” platform theme.
- Improved reliability and scalability.
- Ease of integration.

Factors Fostering Adoption Of Mobile Cloud Computing

- **Trends and demands:** Customers expect convenience in using companies' websites or applications from anywhere and at any time. Mobile Cloud computing is meant for this purpose. Users always want to access business applications from anywhere, so that they can increase their productivity, even when they are on the commute.
- **Improved and increased broadband coverage:** 3G and 4G along with WiFi, femtocells, are providing better connectivity for mobile devices cloud computing.
- **Enabling technologies:** HTML5, CSS3, a hypervisor for mobile devices, cloudlets and Web 4.0 are enabling technologies that will drive adoption of mobile cloud computing.

Characteristics Of Mobile Cloud Computing Application

- **Cloud infrastructure**: Cloud infrastructure is a specific form of information architecture that is used to store data.
- **Data cache**: In this, the data can be locally cached.
- **User Accommodation**: Scope of accommodating different user requirements in cloud app development is available in mobile Cloud Computing.
- **Easy Access**: It is easily accessed from desktop or mobile devices alike.
- Cloud Apps facilitate to provide access to a whole new range of services.

Mobile Cloud Computing Working

- On a remote data center, Mobile Cloud Applications are operated generally by a third-party, data is stored, and compute cycles are carried out. The uptime, integration, and security aspects are taken care of, by a backend, which also enables support to a multitude of access methods.
- These apps can function online quite well, however, they need timely updating.
- These need not be permanently stored on the device but they do not always occupy any storage space on a computer or communications device.

Benefits of Mobile Cloud Computing

- Mobile Cloud Computing saves Business money.
- Because of the portability which makes their work easy and efficient.
- Cloud consumers explore more features on their mobile phones.
- Developers reach greater markets through mobile cloud web services.
- More network providers can join up in this field.

Challenges of Mobile Cloud Computing

- **Low bandwidth:** This is one of the big issues in mobile cloud computing. Mobile cloud use radio waves which are limited as compared to wired networks. Available wavelength is distributed in different mobile devices. Therefore, it has been three times slower in accessing speed as compared to a wired network.
- **Security and Privacy:** It is difficult to identify and manage threats on mobile devices as compared to desktop devices because in a wireless network there are more chances of the absence of the information from the network.
- **Service Availability:** Users often find complaints like a breakdown of the network, transportation crowding, out of coverage, etc. Sometimes customers get a low-frequency signal, which affects the access speed and storage facility.

Mobile database

- A Mobile database is a database that can be connected to a mobile computing device over a mobile network (or wireless network).
- Here the client and the server have wireless connections.
- In today's world, mobile computing is growing very rapidly, and it is huge potential in the field of the database. It will be applicable on different-different devices like android based mobile databases, iOS based mobile databases, etc. Common examples of databases are Couch base Lite, Object Box, etc.

Features of Mobile database :

- A cache is maintained to hold frequent and transactions so that they are not lost due to connection failure.
- As the use of laptops, mobile and PDAs is increasing to reside in the mobile system.
- Mobile databases are physically separate from the central database server.
- Mobile databases resided on mobile devices.
- Mobile databases are capable of communicating with a central database server or other mobile clients from remote sites.
- With the help of a mobile database, mobile users must be able to work without a wireless connection due to poor or even non-existent connections (disconnected).
- A mobile database is used to analyze and manipulate data on mobile devices.

Mobile Database typically involves three parties :

- **Fixed Hosts –**
It performs the transactions and data management functions with the help of database servers.
- **Mobiles Units –**
These are portable computers that move around a geographical region that includes the cellular network that these units use to communicate to base stations.
- **Base Stations –**
These are two-way radios installation in fixed locations, that pass communication with the mobile units to and from the fixed hosts.

Limitations :

- It has Limited wireless bandwidth.
- In the mobile database, Wireless communication speed.
- It required Unlimited battery power to access.
- It is Less secured.
- It is Hard to make theft-proof.

Mobile Ad hoc Network (MANET)

- MANET stands for Mobile Adhoc Network also called a wireless Adhoc network or Adhoc wireless network that usually has a routable networking environment on top of a Link Layer ad hoc network.
- They consist of a set of mobile nodes connected wirelessly in a self-configured, self-healing network without having a fixed infrastructure.
- MANET nodes are free to move randomly as the network topology changes frequently. Each node behaves as a router as they forward traffic to other specified nodes in the network.

- 
- A temporary variety of local area network is an ad hoc network (LAN). An ad hoc network turns into a LAN when it is permanently installed.
 - An ad hoc network may accommodate multiple users at once, although performance may suffer.
 - As long as the hosting device has internet connectivity, users can also use an ad hoc network to connect to the internet.

- 
- Security is one of the main issues with an ad hoc network. Cybercriminals can typically connect to a wireless ad hoc network and, consequently, to the device if they come within signal range.
 - The network name cannot be disguised if users are in a public area since users cannot prevent their ID broadcast in ad hoc mode.
 - Ad hoc networks are not always appropriate because of this.
 - Ad hoc connections, however transitory and only reachable within 100 metres, can be useful in some circumstances. Attackers are unable to access a gadget from a distance and are limited in the amount of time they have to plot their attack.

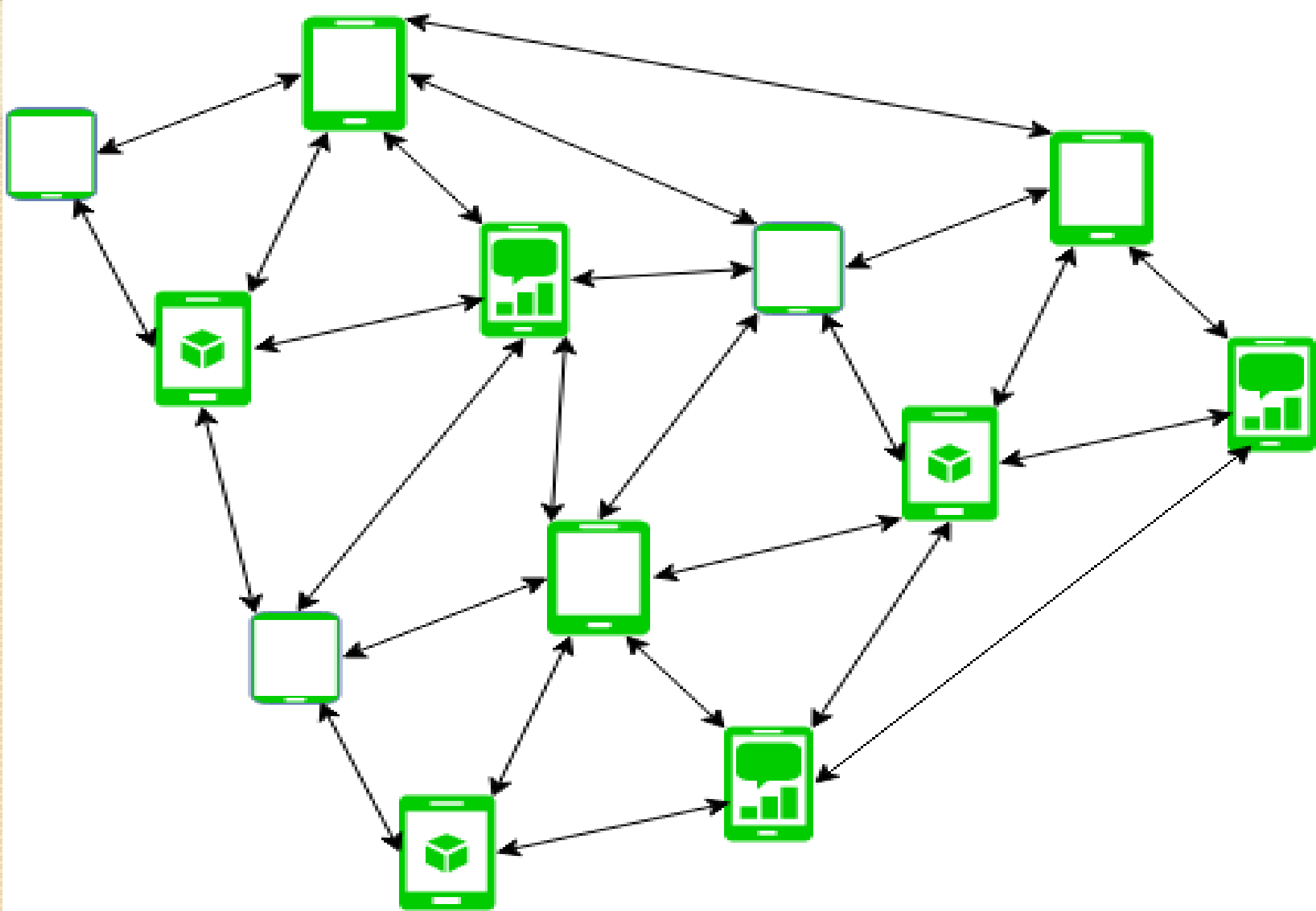


Figure - Mobile Ad Hoc Network

Characteristics of MANET –

- **Dynamic Topologies:**
Network topology which is typically multihop may change randomly and rapidly with time, it can form unidirectional or bi-directional links.
- **Bandwidth constrained, variable capacity links:**
Wireless links usually have lower reliability, efficiency, stability, and capacity as compared to a wired network
- **Autonomous Behavior:**
Each node can act as a host and router, which shows its autonomous behavior

Characteristics of MANET –

● **Energy Constrained Operation:**

As some or all the nodes rely on batteries or other exhaustible means for their energy. Mobile nodes are characterized by less memory, power, and lightweight features.

● **Limited Security:**

Wireless networks are more prone to security threats. A centralized firewall is absent due to the distributed nature of the operation for security, routing, and host configuration.

● **Less Human Intervention:**

They require minimum human intervention to configure the network, therefore they are dynamically autonomous in nature.

Pros and Cons of MANET –

- **Pros:**
- Separation from central network administration.
- Each node can play both the roles ie. of router and host showing autonomous nature.
- Self-configuring and self-healing nodes do not require human intervention.
- Highly scalable and suits the expansion of more network hub.

Cons:

- Resources are limited due to various constraints like noise, interference conditions, etc.
- Lack of authorization facilities.
- More prone to attacks due to limited physical security.
- High latency i.e. There is a huge delay in the transfer of data between two sleeping nodes.

MOBILE COMPUTING OPERATING SYSTEM

- An operating system is necessary for mobile computing. Without taking into account the capabilities and specifications of the hardware, an operating system (OS) enables the user to run an application.
- Additionally, it offers tools for organizing various tasks into a system's schedule. Even the personal information manager (PIM) and APIs for using SMS, MMS, GPS, and other apps are provided by an OS.
- Tasks and memory can be managed by an OS using tools including creation, activation, deletion, suspension, and delay. It offers the features necessary for the system's many tasks to be synchronized. There may be several threads for a task. It offers thread synchronization and priority distribution.

- Additionally, an OS offers interface for software at the application layer, middleware layers, and hardware devices to communicate with one another. It makes it easier for software components to run on a variety of hardware.
- For the device's graphic user interface (GUI), an OS offers programmable libraries.
- Many user-operated devices require User application's GUIs, Voice User Interface (VUI) components, and phone Application Programming Interface (API).
- Device drivers for USB, keyboard, displays, and other devices are also provided by an OS. Middleware, applications, and a fresh environment for developing applications are all provided by mobile OS.

Brief Details about Database management system

- A database management system is used for organizing information that can be easily persisted, queried and managed. Based on the nature of the data, we can leverage various kinds of databases. Data is organized in structured way to easily persist and retrieve in relational database systems.
- In No-SQL databases we mainly store the key-value pairs, or documents or graph-based data or unstructured data.

Operation

- We can perform various operations on the data stored in the database. We have given the key operations:
- • Query operation – we can query the structured data from the database table. We can also filter the data using a conditional operator.
- • Insert operation – We can insert the data into the specific table using the insert operation.
- • Delete operation – We can delete a row(s) from the database based on the condition.

Types of databases

- There are various kinds of databases. We can use the most appropriate database based on the use case. Given below are the some of the most popular databases
- - Relational database: The data is stored in tables which are related to each other. For structured data use cases such as financial data we use relational database. MySQL, Oracle, MS SQL Server are some of the popular relational database systems.
- - No SQL database: The database schema/model is flexible in No SQL database. We can store the session data, shopping card data in No SQL database. MongoDB, DocumentDB are few examples of No SQL database engines.
- - XML Database: We use the XML tags as the keys in the XML database. We can use the XML database to manage the key value pair use cases.

Mobile Database:

- A Mobile Database is a type of database that can be accessed by a mobile network and connected to a mobile computing device (or wireless network).
- Here, there is a wireless connection between the client and the server. In the modern world, **Mobile Cloud Computing** is expanding quickly and has enormous potential for the database industry.
- It will work with a variety of various devices, including Mobile Databases powered by iOS and Android, among others. Couchbase Lite, Object Box, and other popular databases are examples of databases.

Mobile database management systems

- SQL Anywhere mobile device database, IBM DB2 Everyplace, Oracle 9i Lite, Microsoft SQL compact, SQLite are some of the popular mobile device databases.
- All mobile database systems use small memory footprint due to the limited resource availability on mobile devices. Mobile devices can store the application configuration, user preferences, user profile and other offline data in the local database.

Mobile Database Environment

Components:

- For storing the corporate and providing the corporate applications, a Corporate Database Server and DBMS is used.
- For storing the mobile data and providing the mobile application, a Remote Database and server are used.
- There is always a two-way communication link present between the Mobile DBMS and Corporate DBMS.

Features of Mobile Database:

- There are a lot of features of Mobile Database which are discussed below:
- As more people utilize laptops, smartphones, and PDAs to live on the go.
- To prevent frequent transactions from being missed due to connection failure, a cache is kept.
- Mobile Databases and the main database server are physically independent.

Features of Mobile Database:

- Mobile gadgets hosted Mobile Databases.
- Mobile Databases can communicate with other mobile clients or a centralized database server from distant locations.
- Due to unreliable or nonexistent connections, mobile users need to be able to operate without a wireless connection with the aid of a Mobile Database (disconnected).
- Information on mobile devices is analyzed and managed using a Mobile Database.

- 
- Mobile Database Consists of **Three Parties** Which are Described Below:
 - **Fixed Hosts:**
With the aid of database servers, it handles transactions and manages data.
 - **Mobile Units:**
These are mobile, transportable computers, and the cell tower they utilize to connect to base stations is a part of that geographical area.
 - **Base Stations:**
These two-way radios, which are installed in fixed places, allow communication between the stationary hosts and the mobile units.

- 
- In many instances, a user may utilize a mobile device to log in to any corporate database server and deal with data there, depending on the specific requirements of mobile applications.
 - While in other cases, the user can upload data collected at the remote location to the company database or download it and work with it on a mobile device.
 - The interaction between the corporate and mobile databases is frequently intermittent and only occasionally establishes or establishes a link for a brief period of time.

- **Limitations:**

- There are a lot of limitations or drawbacks available, which are pointed out below:
- Its wireless bandwidth is restricted.
- It is very difficult work to make this database theft-proof.
- To operate this, we need unlimited battery power.
- Wireless communication speed suffers in mobile databases.
- In terms of security, it is less secure.

Mobile Device Data Store Methods

- Mobile devices have limited storage capacity. We cannot store large quantities of data on mobile devices.
- For dynamic mobile apps we need to retrieve the data from the server. If the mobile device tries to retrieve large quantity of data for every screen refresh, it impacts the end user experience.
- Hence to balance the dynamic nature of the data and the mobile app performance, the mobile app needs to store the data locally on the device.
- Storing the frequently used data locally on the device is referred to as caching or hoarding

Single Tier Architecture

- In this architecture model, the mobile app directly interacts with the embedded database on the mobile device.
- We could use embedded mobile database such as SQLite which stores the frequently access mobile app data. The embedded mobile database stores the core application data such as business data, location data, user data that enables the mobile app to be operated in the offline mode as well.
- In single tier mobile database architecture, the entire mobile app data is hoarded within mobile device.

Multi-Tier Architecture

- In a typical multi-tier architecture, the mobile app uses the data stored in the remote database through APIs.
- Figure 2 depicts a multi-tier architecture. The mobile app in tier 1 invokes the business services in tier 2 which gets the data from the database in tier 3. The mobile app can optionally cache the retrieved data locally in a cache or a database. This reduces the overall latency in invoking the data from the remote database. We have depicted the local caching architecture in Figure 3.



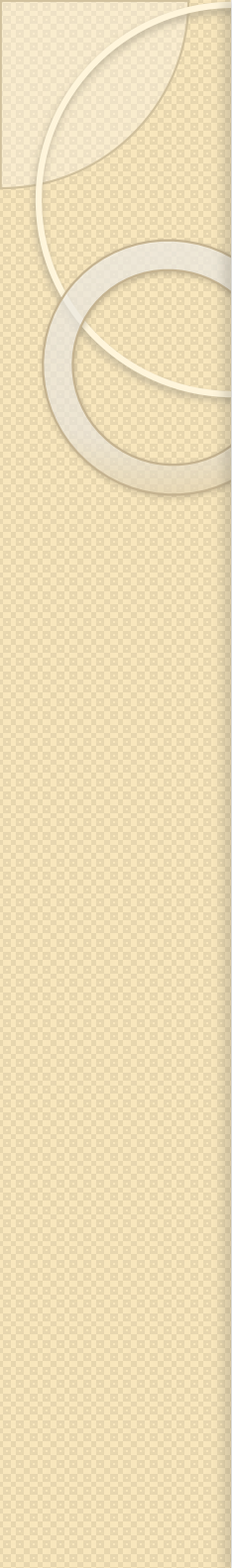
● **Caching Patterns**

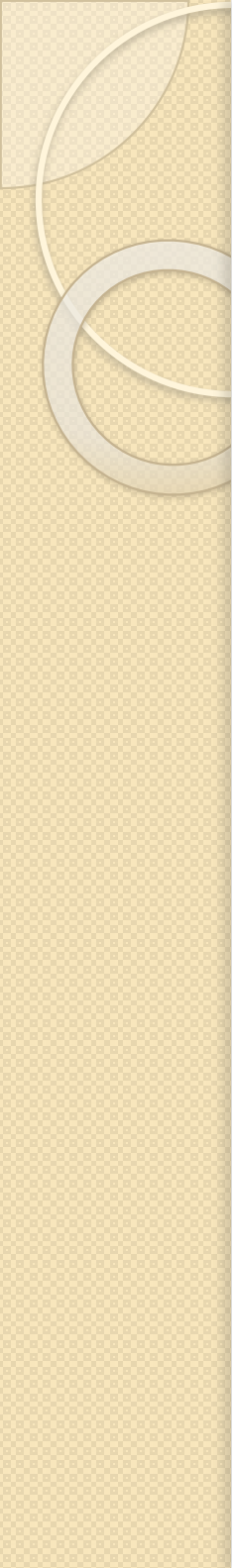
- In the multi-tier architecture the mobile app caches the data to reduce the overall latency.
- If the data is regularly accessed (for instance user detail information or static information that is displayed on each mobile screen) then it is an ideal cache candidate.

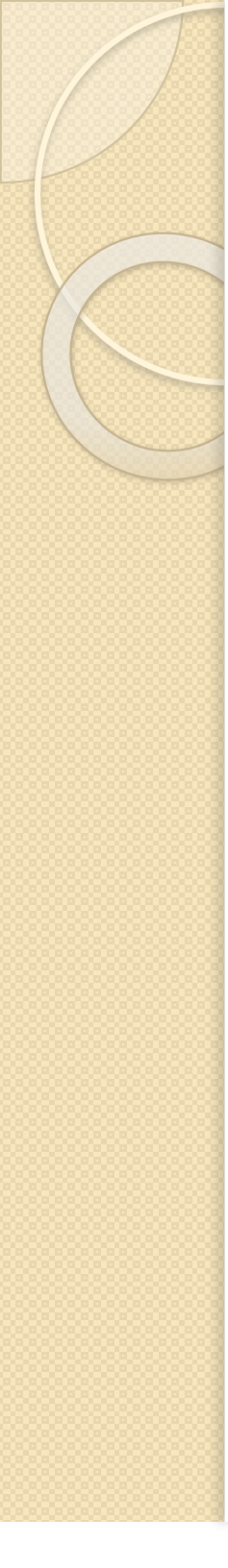
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- Broadly there are two methods to populate the cache.
 - In the first method the mobile app pulls the required data from the remote server and caches that data.
 - The mobile app can pull the most frequently used data to reduce the overall latency.
 - The second one is the push method where the server pushes the data.
 - The push is mainly used by the server to synchronize the server data with the client data.
 - Among the push and pull methods, the pull method is optimal as only the required data is pulled when it is required.

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- 
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CLIENT SERVER

COMPUTING USING MOBILE

- A distributed computing architecture known as client-server computing uses servers and clients as its two types of nodes.
- The server can be asked by a client for information or responses, which the client can subsequently employ in calculations.
- The client can cache these records on the client device or access them directly from the server.
- The data may be accessible at the client's request, via broadcasts, or by server distribution.

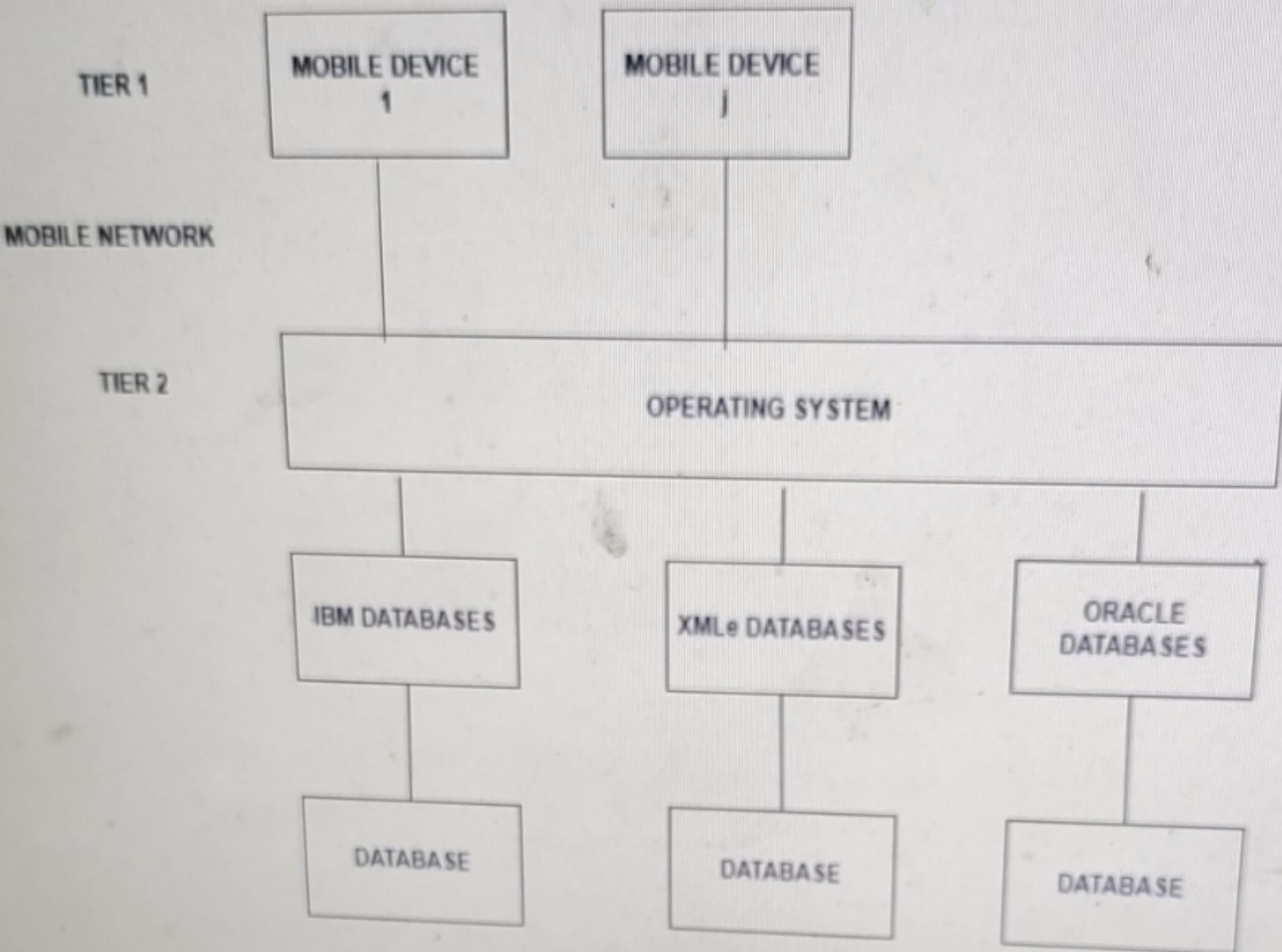
Client-Server Two-tier Architecture

- An application server may be required by a particular application to distribute a local copy of data to numerous devices.
- On the devices, the application can now execute independently without the need for run-time retrieval. Through a synchronization API, the multiple APIs communicate with one another.
- When records are amended at the server end, synchronization means that the cached copies on the client devices should also be updated.
- Since different devices may use different platforms, the APIs are created as independently of hardware and software platforms as possible.

Three-tier Architecture for Mobile Computing

- In three-tier computer architecture, the application interface, functional logic, and database are maintained at three distinct tiers.
- Through a synchronization-cum-application server at tier 2, data records at tier 3 are transmitted to layer 1 of the data chain. With the help of business logic, the synchronization-cum application server's synchronization and server programs retrieve data records from the enterprise layer (tier 3).
- Using a connectivity protocol, the enterprise tier connects to the databases and transfers the database records to tier 2 in accordance with the business logic query

- 
- Additionally, a three-tiered data hoarding system is displayed. A middle server known as the synchronization server transmits and synchronizes copies to numerous mobile devices. For the applications 1 to j, local copies of the databases 1 to l are stored on the mobile devices.
 - In the image, an enterprise database connection to a synchronization server is also shown.
 - This server synchronizes the enterprise database server with the local copies needed by the apps. In the event of a Web-client at tier 1, HTTP can also be utilized as a synchronization protocol. Internet-based communication from Tier 1 to Tier 2 allows for further connectivity to be created using HTTP or HTTPS.



DESIGN CONSIDERATIONS FOR MOBILE COMPUTING

- **Considerations for Frameworks and Programming Languages:**
- The mobile computing architecture employs a number of programming languages. Java is one of the most used languages for mobile computing.
- This is due to Java's most significant feature, platform independence, which means that Java program codes are independent of the CPU and OS utilized in a system.
- This results from a typical compilation into bytes. J2SE is the name of Java 2's standard edition. There are two editions with a small memory footprint: Java2 Micro edition (J2ME) and Java Card (Java for smart card).
- These are the two most popular languages for mobile computing and creating apps for platforms used by mobile devices

- 
- Mobile service apps that are Web- and enterprise server-based employ Java 2 Enterprise Edition (J2EE). The other two widely used programming languages are C# and JavaScript.
 - Program compilation for these languages depends on the CPU and operating system being utilized. The benefit of C# is that it makes it possible to build UIs, and JavaScript makes it possible to create contents on the fly.
 - Server communications and client messages are presented using the markup language XML HTML5.

- 
- Popular mobile application frameworks like Python 2.7 and 3.0 support scripting languages like JavaScript as well as programming languages like Java or C#. A framework for creating Internet and Web applications is provided by QtScript. For enterprise applications, J2EE is used, and for web pages, JSP.
 - DotNet offers a platform for creating apps, web applications, and Internet services. For enterprise applications, it uses Visual Studio and ASP.NET, while for web pages, it uses JSP.

Operating System Considerations:

When designing an app, an OS that is compatible with the mobile device is taken into account.

The user can launch an application on the iPhone 6. Thanks to iOS 8, which is used by an app designer for an iPhone app.

The following are some examples of how an app creator makes advantage of the OS's functions:

- 
- Hardware features and specs
 - Supporting the creation, activation, deletion, suspension, and delay of threads and the scheduling of numerous threads in a system
 - Interfaces for exchanging information between hardware devices, middleware layers, and application layer software components
 - Facilitating the hardware's ability to run software components
 - Adjusting the device's library settings for the GUI and VUI components
 - Artistic writing, music composition, and graphic design

- 
- **Middleware functions:**
 - Middleware is a term used to describe the computer programmes that connect application components to network-distributed components.
 - Between a user application and an operating system, middleware functions as additional software. A
 - Additionally, Mobile OS offers middleware components. The following uses are made of middleware applications:

- 
- a) In order to find adjacent Bluetooth devices
 - b) To find the hotspot nearby
 - c) To synchronize a device with a server or an enterprise server
 - d) To retrieve data from a network database, which may be in Oracle or DB2 format
 - e) For transaction processing
 - f) To use an application server for a stateless Internet session
 - g) For service discovery
 - h) For application platform and service availability adaption

- 
- **Data Synchronization and Dissemination:**
 - A mobile phone can be used as a data access device to access the server of the service provider and retrieve information.
 - Smartphones serve as enterprise data access devices in networks used by businesses. The data is distributed to the enterprise mobile device, such as the BlackBerry handset, by an enterprise server.
 - A data access device for accessing music or videos is something like an iPod, iPhone, iPad, or tablet. Files can be downloaded via a link and saved or played afterward. These days, students can record lectures from professors and access e-learning materials using an iPod, iPhone, iPad, or tablet.

- 
- The three ways for disseminating data are:
 - (i) broadcasting or pushing (such as sending unwanted SMSs to mobile phones),
 - (ii) pulling (such as downloading a ringtone from the mobile service provider), and
 - (iii) a push-pull hybrid.
 - **Podcasting** is a modern type of broadcasting. It is a brand-new technique for sharing multimedia files. For instance, files for music videos and audio programs are disseminated online in formats that can be played on computers and mobile devices.

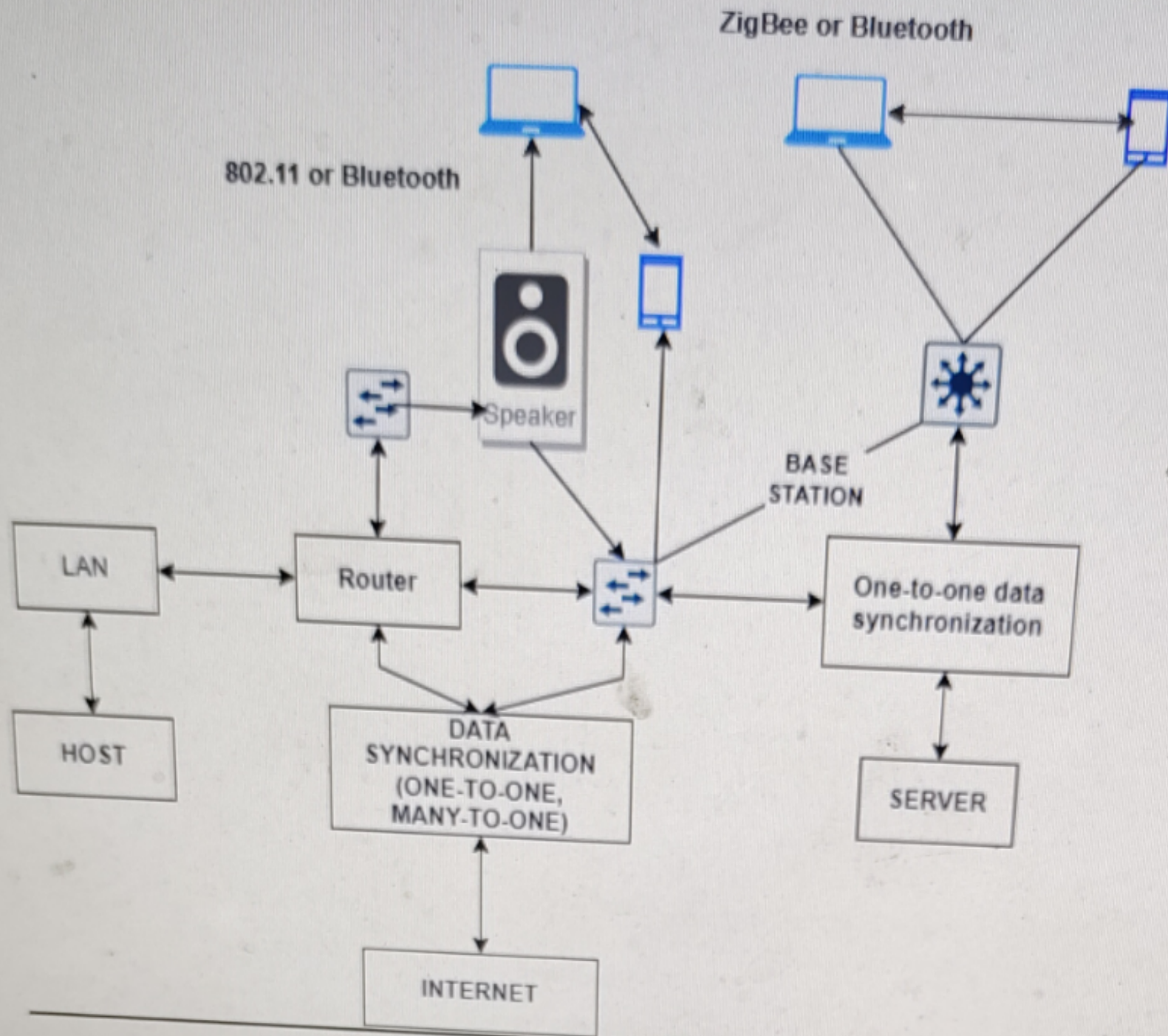
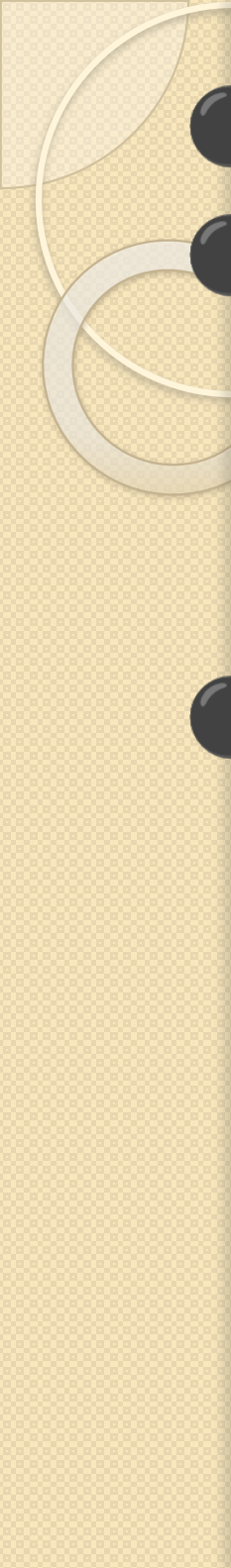


Fig 6: Data synchronization paths



● Synchronization

- Data synchronization, often known as synchronization, is the process of updating data across many databases so that each repository includes the same data.
- Data synchronization ensures that all servers of a mobile service provider have the same selection of ringtones if, for instance, a new, well-liked ringtone is added to one of the servers of the service provider. Furthermore, any new data should be made available to all the devices linked to the server.

- 
- **One-to-one synchronization:** When two ends of a data stream are in sync, every modification made at one end is immediately reflected at the other.
 - The information on the two ends should be the same or consistent. In a data-synchronized enterprise network, the change is reflected at the enterprise servers, for instance, if the address of a mobile device user changes and the user enters the new address in the device's address book.
 - In this situation, a GSM or CDMA network is used to synchronize devices. But when an iPod is put next to a PC and iTunes is downloaded into the PC, the synchronization takes place via Bluetooth or USB connectivity.

MOBILE COMPUTING AND THE APPS

- Utilizing Mobile IP or WAP gateways of direct HTTP applications over 3G data services, users can access the internet for computing.
- A comprehensive suite called Mobile Enterprise Application Platform (MEAP) is utilized for mobile computing via the Internet. It makes it possible to create mobile applications, which need Web services, servers, and enterprise servers.
- A mobile middleware server, application software, and a mobile client API make up a MEAP system. Uses for a middleware server include:
 - Administration of back-end servers' or systems' data
 - Setting up the app for the mobile device
 - System integration
 - Scalability

- 
- . Communications
 - 6. Support across platforms
 - 7. Security

Mobile web applications and modern web applications: HTML5 is utilised for existing applications that are mobile-friendly and relate to web pages and web forms.

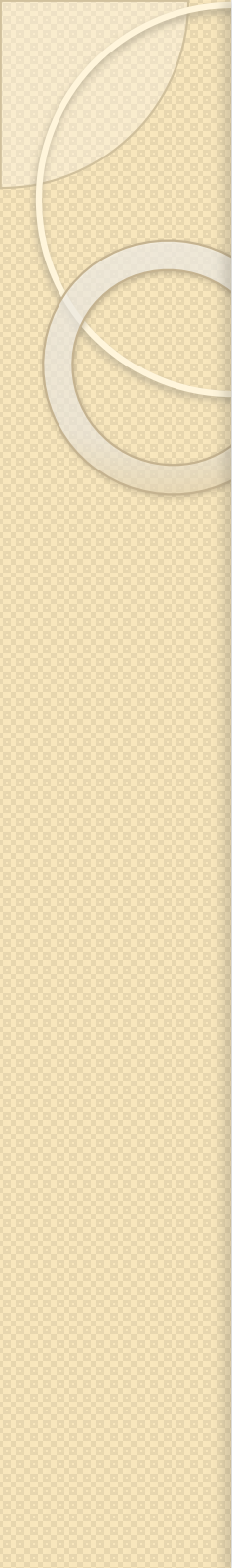
Numerous capabilities and mobile-friendly controls are available with Microsoft ASP. Net 2.0. ASP.NET MVC tools are used to make the current Web applications mobile-friendly (MVC stands for model, view, and control).

NOVEL APPLICATIONS OF MOBILE COMPUTING

• Smartphones A Smartphone is a mobile phone with added processing capabilities that allow for several applications to be used. For instance, Research in Motion, Inc.'s BlackBerry 8530 curve includes additional computing capabilities that make it possible to use the following applications:

- 1. Phone, email, address book, MMS (multimedia messaging service), and SMS (short message service). Web surfing, a calendar, a to-do list, and a memo pad.
- 2. Support for well-liked Personal Information Management (PIM) applications
- 3. Viewing integrated attachments
- 4. QWERTY-style keyboard with Sure Type technology (a computer keyboard has keys in order of Q,W,E,R,T,Y,...)
- 5. Keyboard controls for Send and End
- 6. Use a headset, vehicle buds, or car kits with Bluetooth to talk hands-free.
- 7. When EV-DO compatibility is enabled, the device can be used as a wireless modem for a laptop or personal computer.
- 8. Speaker telephone

- **SmartWatch** The development of mobile computing, medical device design, smart sensor design, and GPS technology led to the creation of the Galaxy Gear S, an innovative application product that Samsung released in August 2014. It is a brand-new smartwatch. It is a brand-new smartwatch.
- These are some of its characteristics: It sports a curved 2-inch display. It offers the capacity to make phone calls without using a smartphone at all. It has Bluetooth and Wi-Fi connectivity options. It permits GPS.
- With its heart rate and UV sensors, it features the S Health app, which alerts the wearer when it's time to eat, when they've had enough activity, and when it's time to relax. It contains walking navigational features.
- It allows watch owners to send a text message from their wrist. Apple Watch provides apps for those who want to lead an active lifestyle, such as Nike+ Running, which allows users to log morning or evening runs.

- 
- **Music, Video, and e-Books** Apps that let you download music, film, and books are available for tablets and other mobile devices.
 - Reading one's favorite books whenever and wherever one wants is now accessible thanks to tablets. AAC-LC plays stereo audio in .m4v, .mp3, and .mov file formats at up to 160 kbps and 48 kHz.
 - Apps for tablets and other devices provide mobile TV and MPEG-4 video. E-books can be read as PDFs or PNGs.



● **Mobile Cheque and Mobile Wallet**

- A mobile-based payment method used when making a purchase is called a "mobile cheque" (mcheque).
- The customer, a certain retail location, and the mobile service provider all text messages to activate the service.
- In order to transfer money to the retailer account, the service provider authenticates the consumer and activates the customer account.
- The method that payments are made for purchases is changing as a result of these mobile devices. Customers are no longer required to have credit cards in their wallets while they shop.
- A mobile wallet is an application where you may access your money via your service provider's account. A bank is used to link the payment when a mobile device connects to the service.

- 
- **Mobile Commerce** The following is an illustration of mobile commerce (m-commerce). Stock quotes can be accessed on demand or in real time using mobile devices.
 - The stock purchaser or seller first sends an SMS for the trading request, then the stock trading service responds in the same manner, requesting authentication.
 - The client delivers the user ID and password through SMS. The customer is then instructed to continue by way of a confirmation SMS. The client sends an SMS to request a certain stock trade.
 - At the stock exchange terminal, the service provider places the order. Within a couple of minutes, the procedure is finished online.

- 
- Mobile-based Supply Chain Management Apps for managing the supply chain on mobile devices effectively manage manufacturing, supply, logistics, warehouse, and other tasks while on the go. The following example provides the clearest explanation of the supply chain management issue.
 - Chocolates cannot be sold by distributors unless they are manufactured. Without a distributor order, the manufacturer cannot produce the chocolates. The supply chain management problem is the name given to this producer-consumer issue.
 - Leading IT firms have created supply chain management system software for mobile devices. Such mobile devices are used by the sales force and the manufacturing facilities to maintain the supply chain



Figure 5 Pull pattern of caching

Another caching technique is the pre-fetch wherein the mobile app prefetches the data in advance in anticipation of its need.

For instance if the mobile user has visited the product list page, the mobile app can prefetch the product details in anticipation of mobile user visiting the product details page.

When the mobile app does not find the required data in the local cache it is called “cache miss” and the mobile app has to go back to the server to get the required data.

We need to use appropriate cache invalidation mechanism to ensure that the mobile client gets the accurate data. There are broadly four cache invalidation methods:

- 
- Stateless asynchronous in which a broadcast message invalidates all cached objects of all clients of the server
 - • Stateless synchronous in which the server broadcasts the cache invalidation method when the data is change or modified.
 - • Stateful asynchronous where the server broadcasts the cache invalidation message only for the specific clients who are impacted. The server does not keep track of the client data
 - • Stateful synchronous where the server keeps track of the client data and broadcasts cache invalidation message to the specific client when the data is changed.

MOBILE INTERNET APPLICATIONS

Markup language will generate page layout , presentation components and enable us to interact with the server.

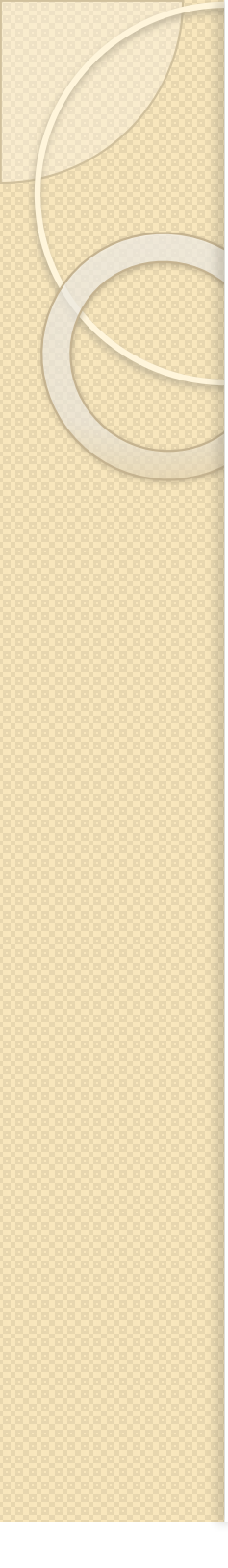
Mobile app is a software that runs on the mobile device that provides the user interface for interaction.

- For web pages the most popular markup language is HTML (hypertext markup language). HTML defines the web page components so that the browsers can render the web page as per the specification.

- HTML also posts the client requests to the server and gets them server response. Mobile applications use XML for managing the data.

- Mobile devices use that networks like 3G 4G 5G and protocols like wireless application protocol (WAP) to connect to the internet. The connecting devices use xml-based wireless markup language (WML) for exchanging information.

- 
- **Mobile Application development**
 - Mobile application provides the mobile user with a graphical user interface and intuitive interfaces to get the required information.
 - Modern mobile applications use n tier architecture where the mobile app acts as a client and the server is deployed on a remote machine. The server managers a huge database and provides API to expose the data to the mobile client.
 - There are primarily two main development platforms for mobile app: iOS and Android. Android mobile apps can be developed in Java and the iOS mobile apps are developed in frameworks such as Swift UI and UIKit.



Mobile app provides the below given functionality:

1. Creation of a graphical user interface
2. Allowing the end user to use the gestures, clicks to interact with application
3. Interface with remote service to get the data on demand.

INTRODUCTION TO XML

- Extensible markup language (XML) is a declarative markup language uses standard set of tag and attributes for the presenting the data.
- The tag defines The high-level element entity and attributes define the properties of the entity.
- There are multiple XML based standards that's popular for data exchange and data transmission. XML is a platform and language independent structure format.

- 
- XML format is used in variety of scenarios.
 - Firstly XML helps us to manage the data in structured way using well-defined tags and attributes.
 - We can also define the information rules and constraints.
 - XML is also used to share the data across two systems using standard formats.
 - Servers can push the data to the mobile clients using XML data format. We can also represent the object relationship and hierarchies in XML format.

- 
- XML can be used as a database where we manage the information and XML can also be used to specify the commands, define the page layout, specify the constraints, enforce the rules, establish the relationships between entities.
 - As XML is extensible, we can also add custom tags, custom attributes and elements as part of the XML

